

From Fog to Smog: The Value of Pollution Information[†]

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In 2013, China launched a landmark program to monitor air quality and disclose real-time data, significantly increasing the public's access to and awareness of pollution information. The program triggered cascading behavioral changes such as stronger avoidance of outdoor pollution exposure and increased spending on protective products. These behavioral responses mitigated the mortality impact of air pollution. Conservative estimates indicate that the program's health benefits outweigh the costs by an order of magnitude. The findings highlight the benefits of improving public access to pollution information in developing countries which often experience severe air pollution but lack pollution data collection and dissemination. (JEL D83, D91, I12, O13, P28, Q51, Q53)

While many developing countries are experiencing some of the worst pollution in the world, residents in these countries lack access to credible and readily available pollution information. This information is either not collected, deliberately kept secret, or intentionally distorted by their governments.¹ Although economists have long emphasized the importance of information in decision-making (Stigler 1961), little is known about the effects of collecting and disseminating pollution information on the daily lives of citizens in developing countries. Filling this knowledge gap is important, particularly because public funding for improving information

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¹In 2018, among the 20 countries with the worst fine particulate matter (PM_{2.5}) pollution (with annual median concentration greater than 46 ug/m³), only four (Nepal, Saudi Arabia, India, and China) have installed a pollution monitoring system.

infrastructure in developing countries competes with meeting basic needs in health care, nutrition, and education.

China provides a perfect setting for studying the value of pollution information, both because of the high levels of air pollution it has experienced and because of the sharp and sudden change of its policy to monitor and disclose real-time air quality information. During the 2000s, daily average concentrations of fine particulate matter ($PM_{2.5}$) in China is estimated to exceed $50 \mu g/m^3$, five times above the World Health Organization guideline. Despite the hazardous pollution levels, a comprehensive monitoring network did not exist. Public dissemination of the scant pollution data that were collected was restricted and controlled. As a result, there was widespread confusion and obfuscation among government agencies and the general public about whether fog or severe air pollution caused murky skies.² In 2013, against the backdrop of public concerns over the lack of transparency, China made a dramatic change: it launched a nationwide program that began monitoring air quality and disclosing the information to the public in real time (henceforth, the monitoring program). This landmark change represents an important moment in the history and evolution of China's environmental regulations, and the implementation of the program offers a unique opportunity to study how the availability of real-time information on air pollution levels affect household behavior and individual health.

We present the first empirical analysis of this natural experiment and our analysis proceeds in two steps. First, we document what happens to pollution information availability and citizen awareness once public air quality information became widely available. Second, we study how the monitoring program affects behavioral and health outcomes, focusing on two central objects of the literature: *avoidance behavior* and the *health effects of air pollution*. We measure these objects by examining how outdoor activities and health outcomes respond to short-term pollution fluctuations, which we will henceforth refer to as the pollution "gradients," ($\partial \text{Outdoor Activities} / \partial \text{Pollution}$) and ($\partial \text{Health} / \partial \text{Pollution}$). Our empirical focus is to estimate how the monitoring program changed these gradients. The basis of our identification strategy is the staggered introduction of the monitoring program across three waves of cities and completed over the course of two years. The sequence of the staggered rollout was a top-down decision based on administrative hierarchies and characteristics determined long before the program.³ Thus, the rollout schedule was uncorrelated with the day-to-day variation of local pollution levels, as validated in our empirical analysis. In addition, there were no other programs or regulations following a similar rollout schedule during our sample period. Importantly, our identification strategy uses an econometric insight—in particular, two orthogonality conditions that can be tested directly with data—which allows us to causally identify the changes in the two gradients using OLS regressions.

²Both fog and smog could cause low visibility and mostly occur in the fall and winter months, contributing to confusion among the general public. However, fog and smog have different compositions with dramatically different health implications. Fog forms when water vapors under humid conditions condense into water droplets or ice crystals that are suspended in the air near the ground. Smog, a type of intense air pollution, is caused by aerosol particles such as $PM_{2.5}$ that carry toxins and hence impair health when breathed in.

³This includes characteristics such as the location of the city (e.g., whether it is located in a special economic zone), its status as a provincial capital, and any prior designations it may have received (e.g., being designated as an environmental exemplary city in 2007).

We build the most comprehensive database yet compiled on a rich set of outcomes—including social awareness of air pollution, air quality measures from satellites, economic activities, and health outcomes—that covers the period before and after the monitoring program. We first document that the monitoring program has profoundly transformed the landscape of public access to pollution information and dramatically increased households' awareness about pollution issues. The frequency of the publication of air pollution-related articles in *People's Daily*, the government's official newspaper, rose from less than once per week to almost daily. Immediately after the program was launched, the term “smog” (*wu mai* in Chinese) became, for the first time, a buzzword in social media. National surveys indicate that respondents' concern about environmental issues has significantly increased following the program, with the increase in concern being proportionate to local air pollution levels where the respondent resides.

The changes in information access and in public awareness triggered a cascade of behavioral changes. Our analysis takes advantage of the universe of credit card and debit card transactions that occurred at physical stores in China from 2011 to 2016. We measure the extent to which people avoid pollution by analyzing the “shopping-pollution gradient,” which reflects how outdoor shopping trips vary with weekly fluctuations in ambient pollution, and evaluate how the monitoring program impacts this gradient. We show that the monitoring program boosted pollution avoidance by triggering a negative shopping-pollution elasticity of 1.4 percent, a pattern that is robust across a host of econometric specifications. The increase in avoidance concentrates in plausibly “deferrable” consumption categories, including supermarket shopping, restaurant dining, and entertainment. We further examine the impact of the monitoring program on *defensive purchases* using a separate dataset on monthly air purifier sales in 50 large cities in China. We find that the monitoring program substantially intensified defensive spending. Within one year of the program implementation in a city, purchases of air purifiers almost doubled and became highly correlated with contemporaneous local pollution levels.

The evidence indicates that there is a strong demand for public information on air pollution and that households gained new information from the government monitoring program based on the observed changes in behavior. It is natural to ask whether households obtained their own (proxy) information about air pollution in the absence of government-provided monitoring data before the monitoring program. We consider two potential sources of private information. First, we examine whether people can use visibility to infer pollution levels. We show that the relationship between visibility and pollution is noisy, especially when visibility is poor which could arise from either fog or smog. Second, we consider the city of Beijing where the US Embassy has conducted independent pollution monitoring since 2008. The embassy data were released on Twitter, which was (and still is) not accessible in China, so most people do not have easy access to it. Our econometric analyses show that households were not able to effectively engage in pollution avoidance with these private sources of information prior to the monitoring program.

The behavioral changes we document hold the potential to mitigate some of the devastating health consequences of severe air pollution in China. In the final part of the paper, we examine changes in the mortality-pollution relationship that occurred over time as the public's access to information improved. Using nationally representative

mortality data from the Chinese Center for Disease Control and Prevention (CDC), we find a 1.3–2.3 percentage-point reduction in the mortality-pollution elasticity after monitoring began. The effect was concentrated among individuals aged 60 and above, and was more pronounced for deaths due to cardiovascular and respiratory causes than for other causes. As expected, the monitoring program had no effect on injury-related mortality.

Using estimates of the age-adjusted value of statistical life (VSL) and healthcare costs of air pollution in the literature (Ashenfelter and Greenstone 2004; Murphy and Topel 2006; Barwick et al. 2020), we calculate that risk-compensating behavioral responses to the monitoring program generated annual mortality and morbidity benefits of at least 93 billion. Such social benefits outweigh the costs from defensive investments (such as air purifier purchases), avoidance behaviors (such as foregone consumption), and program deployment and maintenance by nearly one order of magnitude, making the monitoring program among the most successful environmental policies documented in a developing country.

We make three main contributions to the literature. First, to the best of our knowledge, our study provides the first empirical estimate of the costs and benefits of a nationwide program on pollution monitoring and information disclosure. In a concurrent study, Greenstone et al. (2022) documented that the automatic data collection underlying this national program improves the data quality and increases online searches for face masks and air filters. Our study complements theirs by showing that the program triggered a cascade of behavioral responses from increased awareness to pollution avoidance, which in turn reduced the health harms of air pollution. Our study is also closely related to the pioneering study by Neidell (2009) on the impact of pollution information on avoidance behavior. In a context where pollution information is widely available via both print and electronic media, the study shows that smog alerts in Southern California reduce attendance in two major outdoor facilities and that the resulting avoidance behavior mitigates the health impact of pollution.⁴ By examining a large-scale information program in China, our empirical findings highlight the considerable benefits of collecting and disseminating pollution information in developing countries, many of which are experiencing the world's worst mortality damage from pollution exposure (Landrigan et al. 2018). The success of China's program provides a benchmark for policy discussions (e.g., cost-benefit analysis) on building information infrastructure in these countries.

Second, our study shows that pollution information is a crucial determinant of avoidance behavior and defensive spending, consistent with findings in Neidell (2009) and Graff Zivin and Neidell (2009). Consumer activities (online searches, day-to-day shopping, and air purifier purchases) exhibit an insignificant response to pollution until such information becomes widely available. This suggests the importance of understanding the degree of consumers' access to information when applying the revealed preference method to estimate the value of nonmarket environmental goods (Gao, Song, and Timmins 2021). To the extent that access to

⁴In the same context, a follow-up study by Graff Zivin and Neidell (2009) highlights that behaviors to avoid pollution risks (e.g., through intertemporal substitution) entails costs. A similar literature quantifies the value of weather forecasts, another type of government-provided information, as an important input to production decisions (Lave 1963; Craft 1998; Shrader 2018; Jagnani et al. 2018).

information is lacking in developing countries, consumers' true willingness to pay for environmental goods could be underestimated. Our findings provide a potential explanation for why environmental quality is severely undervalued in developing countries (Greenstone and Jack 2015), and why the dose-response relationship between pollution and mortality differs across developed and developing economies (Arceo, Hanna, and Oliva 2015).

Third, this study contributes to the broad empirical literature on the role of information in consumer choices. Growing evidence suggests that consumers misperceive product attributes in a wide range of contexts. Examples include food nutritional contents (Bollinger, Leslie, and Sorensen 2011), insurance policy costs (Kling et al. 2012), vehicle fuel economies (Allcott 2013), retirement savings (Bernheim, Fradkin, and Popov 2015), taxation levels (Chetty, Looney, and Kroft 2009), and energy prices (Shin 1985; Ito 2014). Information provision programs can improve consumers' perception of product attributes (Smith and Johnson 1988; Oberholzer-Gee and Mitsunari 2006), change consumer choices (Duflo and Saez 2003; Hastings and Weinstein 2008; Dranove and Jin 2010; Jessoe and Rapson 2014; Newell and Siikamäki 2014; Mastromonaco 2015; Wichman 2017), and drive up average product quality (Jin and Leslie 2003; Bai 2018). In the context of air quality, recent studies have documented behavioral responses to pollution exposure over both the short and long terms. Our analysis shows that these behavioral responses could lead to significantly improved health outcomes at a moderate cost. We use the associated savings in monetary terms to provide a lower-bound estimate of the benefit of pollution information.

The rest of this paper is organized as follows. Section I reviews institutional details on the monitoring program and describes data sources. Section II documents the dramatic changes in information access and awareness after the program. Section III describes the empirical framework. Sections IV and V present the effect of the program on avoidance behavior and on mortality. Section VI calculates the costs and benefits of monitoring program. Section VII concludes.

I. Institutional Background and Data

A. Environmental Regulations

The real-time PM_{2.5} monitoring program that started in 2013 was an important milestone in the history of China's environmental regulations. The program brought about a sharp and sudden change in the access of pollution information and drastically enhanced the public awareness of the health impact of PM_{2.5}. This section aims to help understand the significance of and underlying factors behind the policy change.

A Brief Chronology of Environmental Regulations since 1982.—In 1982, China established its first national ambient air quality standards (NAAQS), which set limits for multiple air pollutants including total suspended particulates (TSP), sulfur dioxide (SO₂), nitrogen oxides (NO₂), carbon monoxide (CO), and ozone (O₃) (Jiang, Li, and Yang 2020). A 1996 amendment expanded the standards to further include coarse particulate matter (PM₁₀). Throughout much of the 1980s, 1990s, and the early 2000s, the primary threat to air quality was considered to be SO₂

due to coal burning. As acid rain caused widespread and visible damage to crops, forests, and the aquatic environment, central environmental regulations focused on the control of acid rain and SO₂ emissions (Yi, Hao, and Tang 2007). The most prominent regulation was the two-control zone policy (TCZ) implemented in 1998 where cities with low pH values of precipitation and high SO₂ concentration were designated as either the acid rain control zone located in southern China, or the SO₂ control zone in northern China. Cities in these control zones were subject to a series of regulatory measures, such as mandatory installation of flue gas desulfurization in coal-fired power plants and closure of small coal-fired power plants (Tanaka 2015). These aggressive policies reduced the average SO₂ concentration by nearly 45 percent from 1990 to 2002, with the majority of cities achieving the national standard by 1998 (Hao and Wang 2005).

Starting in early 2000, the source of air pollution changed from mainly coal burning to a mix of sources, due to declining emissions from coal-fired power plants and skyrocketing emissions from automobiles, manufacturing facilities, and construction. During this period, pollution regulation followed a federalism approach, where the central government established environmental standards that were meant to be met by local government efforts (Ghanem and Zhang 2014). However, this approach proved to be ineffective due to the strong incentives for economic growth at the local level conflicting with weak monitoring and enforcement by the central government—a common issue in developing countries (Greenstone and Hanna 2014; Duflo et al. 2018; Karplus, Zhang, and Almond 2018). Extreme episodes of air pollution were common in many urban areas, particularly during the winter.

In 2012, China's Ministry of Environmental Protection (MEP) revised the NAAQS and, for the first time in history, set the national standards for PM_{2.5}. The new standards were slated to take effect nationwide in 2016. To help achieve the standards, China's State Council released the *Action Plan on Air Pollution Prevention and Control* in September 2013, which set specific targets for PM_{2.5} reduction from 2013 to 2017 and outlined concrete policies such as promoting the role of market-based mechanisms. It also established a monitoring and warning system to cope with severe air pollution events, which is the monitoring program analyzed in this paper.⁵ In March 2014, Premier Li Keqiang described smog as “nature's red-light warning against inefficient and blind development” and declared a “war on smog” at the opening of the annual meeting of People's Congress (Greenstone et al. 2021, 281). For the first time in the history of national five-year plans, the thirteenth Five-Year Plan required cities at the prefecture level or higher to reduce PM_{2.5} concentrations by 18 percent from 2015 to 2020 (Wang 2018).

China's nationwide pollution monitoring and disclosure program, as well as its aggressive policies to reduce PM_{2.5} concentrations marked a significant shift from the country's previous approach of prioritizing economic growth over environmental considerations. Several players and factors likely have worked together in contributing to the dramatic shift. These include a growing body of scientific knowledge on the pernicious effect of PM_{2.5} on human health, independent PM_{2.5} monitoring

⁵ The plan may have been China's most influential environmental policy during the past decade. Under this plan, PM_{2.5} fell by over 37 percent in the Beijing-Tianjin-Hebei Region, 35 percent in the Yangtze River Delta, 26 percent in the Pearl River Delta, and over 30 percent on average in more than 70 major cities (Huang et al. 2018).

conducted by the US Embassy in Beijing which have caused social media storms, and the increasing impacts of environmental NGOs and social media influencers. Other economic factors such as the rapid growth in household income, the emergence of a large middle class, and the corresponding increase in demand for environmental quality, may also have contributed to the shift (Kahn and Zheng 2016). For interested readers, we provide detailed discussions of the policy shift and underlying factors in online Appendix A, which presents a chronology of the key events from June 2004 when “smog” was first known to be used to describe the atmospheric condition to March 2014 when the Chinese Premier Li Keqiang declared the “war on smog.”

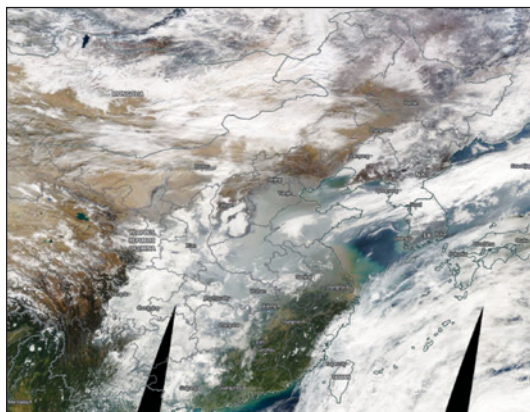
Pollution Information Access and Public Awareness before 2013.—While air pollution has been a long-standing issue, public access to real-time pollution information, especially $PM_{2.5}$ concentration, was nearly absent prior to 2013. The MEP began publishing a daily Air Pollution Index (API) based on reported data on PM_{10} , SO_2 , and NO_2 from local governments for over 40 major cities in 2000.⁶ There are several issues with the API data. First, before 2013, the monitoring stations responsible for collecting the data were run by local governments rather than being directly controlled by the MEP. The environmental bureaus responsible for gathering and reporting the API data were appointed by local governments, which can create conflicts of interest and compromise the reliability of the data. Studies have shown that there has been widespread manipulation of the API data in the past (Andrews 2008; Chen et al. 2012; Ghanem and Zhang 2014). Second, the API did not incorporate data on $PM_{2.5}$, which is believed to be a major contributor to air pollution in China, as the main sources of emissions have shifted from coal-fired power plants to a mix of sources including transportation and manufacturing. Third, the reported data is not real-time, as it is collected manually rather than through automatic process. The daily index covers the 24-hour period from 12 PM of the previous day to 12 PM of the current day, which further contributes to the discrepancy between the reported levels and the public’s perception of air quality.⁷

Without systematic pollution monitoring and reporting, government agencies and the media often characterized high pollution episodes as *fog* rather than *smog*. For example, on November 27, 2011, newspaper headlines and the China Meteorological Administration attributed dense fog in Beijing and northern cities as

⁶The API is a numerical index that ranges from 0 to 500, which is calculated by transforming the concentration of various pollutants (PM_{10} , SO_2 , and NO_2) into a standardized scale. Individual APIs are first calculated for each pollutant, and the overall API is determined by taking the maximum of these pollutant-specific APIs. The “max” pollutant that determines the overall API for the day is called the “dominant pollutant.” According to a study by Chen et al. (2012) that analyzed daily API data from 37 cities between 2000 and 2009, when the API is above 50, PM_{10} is the dominant pollutant in 91 percent of cases, while SO_2 is the dominant pollutant in 8.6 percent of cases. Although the dominant pollutant had shifted from SO_2 to particulate matter (PM) in the 2000s, the government did not systematically collect $PM_{2.5}$ measures before 2013. Therefore, the reported API did not incorporate $PM_{2.5}$, very likely the most harmful pollutant that the public should care about.

⁷Due to the lack of reliable and publicly available air quality information in China, the US Embassy in Beijing installed a rooftop air quality monitor in April 2008 in anticipation of the summer Olympics being held in Beijing. The embassy reported hourly $PM_{2.5}$ levels on its Twitter account to advise US citizens who were traveling or conducting business in China. US consulates in Guangzhou, Shanghai, and Shenyang also followed suit in later years. However, Twitter is not accessible in China, so the data collected by the US Embassy and consulates was not widely circulated among the general public in these cities, even though it has received attention from environmental activists.

Panel A. Satellite picture of the event



Panel B. Satellite-derived pollution levels

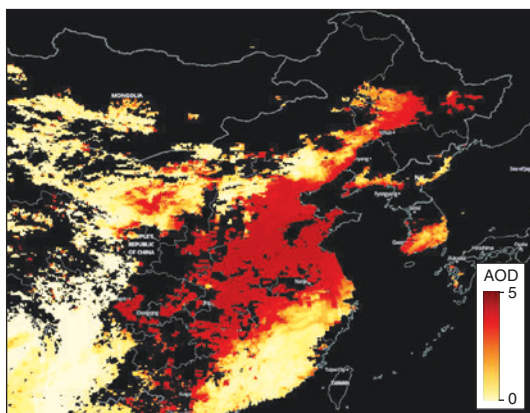


FIGURE 1. CONFUSION BETWEEN HEAVY SMOG AND DENSE FOG BY GOVERNMENT AGENCIES

Notes: This figure illustrates a major air pollution event on November 27, 2011, that was reported by the China Meteorological Administration website with the headline “Dense Fog Causes Traffic Congestion in Many Northern Cities” (https://www.cma.gov.cn/2011xwzx/2011xqxw/2011xqxyw/202110/t20211030_4050034.html, accessed February 5, 2020). Panel A, sourced from NASA, shows the satellite view of China on the same day. Panel B, sourced from NASA MODIS algorithm, shows the satellite-based measure of pollution (aerosol optical depth). Dark red regions have high levels of pollution. Black regions have missing data (with cloud cover or water surfaces).

the reason for widespread flight delays and cancellations. However, the real cause of the low visibility was extreme pollution, as can be seen in Figure 1, which shows the aerosol optical depth (AOD)—a measure of atmospheric particulate pollution obtained through remote sensing, provided by NASA satellites. The average AOD concentration in China is typically around 0.5 (in the United States, the average is usually between 0.1 to 0.15). On that day, the AOD value was 4.5 or higher in many northern cities. A similar pollution event that occurred on December 4 through 6, 2011 was again reported as dense fog by major news media, including China Central Television, the predominant state television broadcaster in Mainland China, and popular websites such as sina.com.

The lack of awareness of PM_{2.5} and the “fog-smog” confusion among the public and the media were reflected upon by the prominent journalist-turned-environmentalist Chai Jing in her high-profile documentary on China’s air pollution titled “*Under the Dome*,” released in February 2015:

... I go back and check the headline from that day’s newspaper (on December first, 2004): ‘Fog at Beijing Capital Airport Causes Worst Flight Delays in Recent Years.’ We all believed that it was fog back then. That’s what we called it ... as a former journalist, I started to blame myself because for all these years I had been reporting stories on pollution all across the country, I always thought pollution was about mining sites and those places near factories spewing smoke plumes. I never thought the skies that we saw every day could be polluted.⁸

The 2013 Monitoring Program.—To effectively monitor local air pollution levels and to address the pitfalls of the previous reporting system (Greenstone et al. 2022), the MEP implemented a nationwide monitoring program starting in 2013. The focus was to build an efficient, scientific system to monitor air quality and to publicly disclose monitoring data in real-time. This monitoring program, the subject of our study, contained two major provisions.

First, it required the installation of a comprehensive network of monitors in 337 cities that in total covered an estimated 98 percent of the country’s population. The program initiated continuous monitoring of all major air pollutants, including PM_{2.5}, PM₁₀, O₃, CO, NO₂, and SO₂. Figure 2 plots the rollout of the program. The rollout was implemented in three waves, and cities are assigned to waves based on administrative hierarchies and designated status that were determined long before the monitoring program. Cities in wave 1 include those in the Jing-Jin-Ji Metropolitan Region, the Yangtze River Delta Economic Zone, the Pearl River Delta Metropolitan Region, direct administered municipalities, and provincial capitals. They are required to start monitoring by December 31, 2012. wave 2 contains cities that were among the Environmental Improvement Priority Cities or National Environmental Protection Exemplary Cities, both designated between 1997 and 2007.⁹ These cities were required to begin monitoring by October 31, 2013. wave 3 includes all remaining cities which were required to monitor by November 30, 2014. When the program was completed, the government had installed more than 1,400 monitoring stations nationwide.¹⁰

There are two nuanced aspects of the program rollout. First, there are a small number of cities that are assigned to waves that depart from the city hierarchy or

⁸The documentary has been compared with Al Gore’s *An Inconvenient Truth* in terms of its style and impact. The film had been viewed over 150 million times on the popular website tencent.com within three days of its release, before it was taken offline. Tu et al. (2020) show that household willingness to pay for clean air increased among those who had viewed the documentary.

⁹The Environmental Improvement Priority Cities were designated in 2007 during the 11th Five-Year period and contained important regional economic centers and cities that faced significant environmental challenges. The National Environmental Protection Exemplary Cities program was created during the ninth Five-Year Plan based on a host of environmental quality criteria.

¹⁰For the list of cities in each wave, see MEP’s official documents “Air Quality Monitoring Program Wave One Implementation Guideline” (May 21, 2012), “Air Quality Monitoring Program Wave Two Implementation Guideline” (March 22, 2013), and “Air Quality Monitoring Program Wave Three Implementation Guideline” (May 7, 2014).

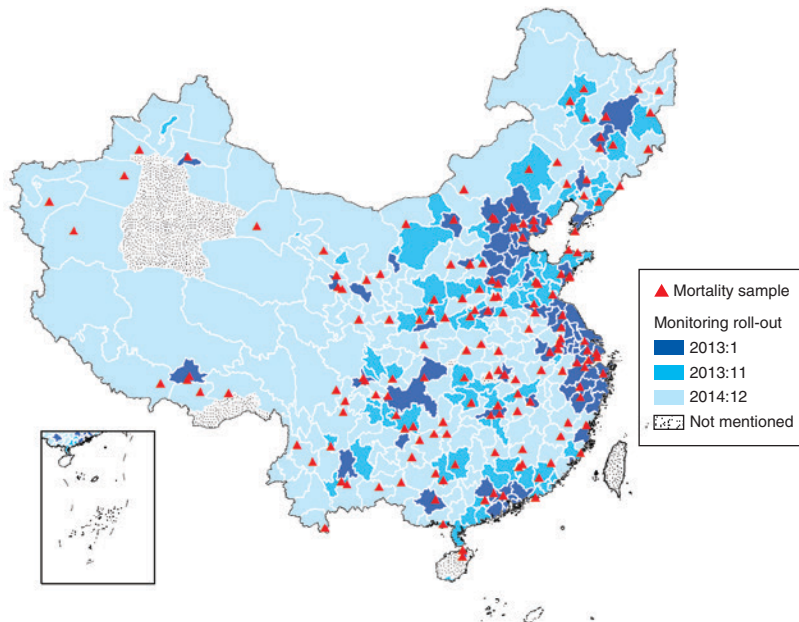


FIGURE 2. MONITORING PROGRAM ROLLOUT AND MORTALITY DATA AVAILABILITY

Notes: This map shows the list of cities by the completion date of the air pollution monitoring and disclosure program. “Not mentioned” are cities where the timing of monitoring is not mentioned in the government’s policy. “Mortality sample” are cities included in the DSP mortality data.

designated status. For example, the city of Xiamen does not belong to any of the three special economic zones, is neither a self-administered municipality nor a provincial capital, but was in wave 1. In Section IIIC, we show that a majority of these deviations are due to geographic contiguity. That is, they border cities in the corresponding wave and it makes sense administratively to group them into one wave. Second, as is often the case in China, the official rollout dates specified in the government guidelines serve as “deadlines” to implement the policy. In practice, cities may finish installing monitors prior to the official rollout dates. We are unaware of reliable data that record the exact dates when cities finished installing monitors. To make progress, we manually search news media and city yearbooks for when pollution information first became available in each city, which we call “field” rollout dates. Our manual data collection effort yields fuzzy information at best. In Section IVD, we show that our findings are robust to using the field rollout dates instead of official rollout dates.

Second, and more importantly, the monitoring network is coupled with a real-time information dissemination system. In contrast to the previous API reporting system, the MEP directly controls the newly established pollution monitors to avoid manipulation by the local government. All monitoring data are automatically streamed to MEP’s information center and broadcast to the general public in real time on the MEP’s Website. Hourly and daily concentrations of $PM_{2.5}$, PM_{10} , O_3 , CO, NO_2 , and SO_2 , as well as an Air Quality Index (AQI) encompassing all six

TABLE 1—SUMMARY STATISTICS

	Cities available	Years available	Frequency	Mean	SD
<i>People's Daily</i> “smog” articles (count per week)	313	2011:1–2016:4	Weekly	0.031	0.310
Baidu web “smog” searches (raw index)	292	2011:1–2016:4	Weekly	285.3	916.5
Expressed environmental concerns (scale 1–10)	126	2012, 2014, 2016, 2018	Panel Survey	5.87	2.69
Air purifier purchases (units per capita)	50	2012:1–2016:4	Monthly	0.067	0.086
Bank card transactions: all (swipes per 10,000 cards)	334	2011:1–2016:4	Weekly	869.1	347.6
Bank card transactions: deferrable (swipes per 10,000 cards)	334	2011:1–2016:4	Weekly	314.4	161.3
Mortality rate: general (deaths per 100,000)	131	2011:1–2016:12	Weekly	10.36	3.46
Mortality rate: cardiorespiratory (deaths per 100,000)	131	2011:1–2016:12	Weekly	5.63	2.58
Aerosol optical depth (pollution index)	334	2006:1–2016:12	Weekly	0.518	0.512

Notes: Except for air purifier purchases, all other datasets are national or nationally representative in scope. The 50 cities with air purifier purchases cover 28 percent of the total population. Self-reported environmental concerns are from the China Family Panel Studies, which is a nationally representative, bi-annual longitudinal survey. Bank card transactions cover all cities and account for 59 percent of national consumption. The 131 cities with mortality data constitute a 5 percent representative sample of the total population. Satellite pollution data covers the entire nation.

pollutants, are available for each individual monitoring station and city.¹¹ Online Appendix Figure E.1 provides a screenshot of the interactive website interface. Importantly, MEP allows private parties to access and stream data directly from its website. This has spurred a surge in media reports and mobile phone applications that display real-time air quality information. Section II provides more detail on the program’s effects on information access and public awareness about PM_{2.5}.

B. Data

We compile multiple datasets to allow for a comprehensive study of the impacts of the monitoring program on a variety of outcomes. The datasets include information on consumer awareness and online search behavior, public opinion toward environmental issues, bank card transactions, air purifier sales, mortality rates, and air quality measures from satellites. Table 1 shows city and time coverage as well as the mean and standard deviation of the primary variables used in our analysis. Our air purifier sales data cover 50 cities or about 28 percent of the national population. We draw public opinions on environmental issues from nationally representative panel surveys that are available for the years 2012, 2014, 2016, and 2018. All other datasets are national or nationally representative in scope and are available at the city-weekly level from January 2011 to April 2016, except for the mortality data which covers a slightly longer time span through December 2016. Below we provide more details on each individual data source.

Mass Media, Mobile Phone Apps, and Web Searches.—The high Internet and mobile phone penetration in China provides a unique opportunity to investigate

¹¹ The AQI is on a scale of 0 to 500 and separated into six color-coded categories, with 0–50 (green) for “good air quality,” 51–100 (yellow) for “moderate pollution,” 101–150 (orange) for “unhealthy for sensitive groups,” 151–200 (red) for “unhealthy,” 201–300 (purple) for “very unhealthy,” and 301 and higher (maroon) for “hazardous.”

pollution awareness using digital sources.¹² We draw on several digital sources to illustrate the evolution of public access to pollution information. First, we look at the publication trends in the *People's Daily*, the government's official newspaper. We pull articles that contain the word "smog" (or "air pollution," or "atmospheric pollution") from the *People's Daily's* digital archive. We also identify the relevant cities mentioned in each of these articles.

Second, we scraped Apple's App Store to obtain release information on Chinese mobile apps that contain air pollution information in December 2015, using keywords including "smog," "air pollution," and "atmospheric pollution."¹³ These apps typically display current hourly pollution levels; some also provide health-related guidelines (e.g., avoiding outdoor activities) when pollution levels are high. online Appendix Figure E.2 provides a sketch of what a typical smartphone pollution app looks like. We also scraped information on apps in other major categories including gaming, reading, and shopping, etc., which will help us tease out differential changes in the availability of pollution apps.

Third, the most widely used search engine in China, Baidu, provides search intensity indices that summarize the number of queries for certain words in a given city and on a given day among both desktop and mobile users since 2011. These search indices are generated using an algorithm similar to Google Trends. We focus on the search index for "smog," the buzzword for air pollution.

Public Opinion.—To complement the media data, we obtain survey data on public opinion from the China Family Panel Studies (CFPS), a nationally representative, bi-annual longitudinal survey of Chinese communities, families, and individuals. CFPS is one of the most widely used datasets to study public finance and labor markets in China, and is often seen as the Chinese equivalent of the Panel Study of Income Dynamics (PSID). Starting in 2012, CFPS asked respondents to state the perceived "seriousness" of a host of social and economic issues (including corruption, education, environment, housing, inequality, jobs, medical care, and social security) on a scale of 1 to 10, with 10 meaning "the most serious." We use the CFPS data from 2012, 2014, 2016, and 2018, which include 83,237 respondents living in 126 cities. Among these cities, 34, 39, and 49 are in the first, second, and third wave of the monitoring program rollout, respectively.

Bank Card Transactions.—We measure shopping trips using the universe of credit card and debit card transactions from UnionPay, the only inter-bank payment clearinghouse in China and the largest such network in the world. The database covers 59 percent of China's national consumption and 22 percent of its GDP in 2015 as shown in online Appendix Figure E.3. For each transaction, we observe the merchant name and location, and transaction amount and time. Online Appendix

¹²Data from the China Internet Network Information Center show that, by the end of 2012, China had about 0.56 billion internet users (or 40 percent of its population). More than 99 percent of internet users have heard of Baidu, the most popular search engine (seconded by Google, 87 percent), and 98 percent have used it in the past six months (seconded by 360.cn, 43 percent). Mobile phone prevalence rose from 73.5 per 100 people in 2011 to 95.6 per 100 people in 2016 (National Bureau of Statistics), with a smart-phone penetration rate of 72 percent in 2013 (Nielsen).

¹³The query returned the 200 most relevant apps for a given keyword.

Figure E.4 shows the spatial distribution of active cards and transactions. Credit and debit cards are widely adopted across the country as the most commonly used transaction method, especially in urban areas (Barwick et al. 2020).¹⁴ To the best of our knowledge, the fine degree of spatial and temporal resolution provided by this dataset represents the most comprehensive data available on consumption activities for China.

Two features of the data are worth mentioning. First, the Point-of-Sale information in the UnionPay data allows us to distinguish between online and offline transactions. Our data contain a small fraction of transactions that are made online, which we drop because tracing these buyers' physical locations is difficult.¹⁵ We focus on the vast majority of the transactions that are made offline to measure *outdoor* purchase trips. Second, we do not observe specific items purchased in each transaction. Instead, UnionPay classifies merchants into over 300 categories, such as department stores and supermarkets. We use the category information in our analysis to measure the general purpose of each trip.

Our key outcome variable is the *purchase frequency per card*, defined as the number of transactions in a city-week per active card, where active cards are cards with positive transactions in a given city-year. We focus on a 1 percent random sample of cards and all of their associated transactions, with an average of 18.3 million active cards at any given point in time.

Air Purifier Sales.—To measure defensive spending, we acquire air purifier sales data from Growth from Knowledge, a leading market research firm, on the total units of air purifiers sold for both residential and institutional purposes. The data are available at monthly frequencies for 50 cities from 2012–2016. Together, these cities account for 28 percent of the national population. Among these 50 cities, 34, 11, and 5 are in the first, second, and third waves of the monitoring program rollout, respectively.

Mortality.—The Chinese Center for Disease Control and Prevention (CDC) operates a Disease Surveillance Points (DSP) system that covers 73 million individuals in 131 cities, a 5 percent representative sample of China's population over the 2011–2016 period. Among these 131 cities, 38 implemented the monitoring program in the first wave, 38 in the second wave, and 55 in the last wave.¹⁶ The DSP mortality database, drawn from hospital records and surveys of the deceased's household, is one of the highest-quality health databases used in recent medical and economic research (Zhou et al., 2016; Ebenstein et al. 2017). We observe the number of persons and total deaths by each city-week-age group, and separately by causes of death.

¹⁴ Since 2016, mobile payment apps including Alipay and Wechat have become popular, but cards transactions still account for more than 50 percent of national consumption.

¹⁵ Online transactions accounted for 5.1 percent of total transaction volume and 3.6 percent by transaction value between 2013 and 2015. The shares were smaller in earlier years.

¹⁶ The original DSP data report information for 161 counties and city districts which are smaller geo-units than cities. To use the same geographic unit of analysis throughout the paper, we aggregate the county/city district level data to the city level.

Air Pollution.—To overcome the challenge that reliable pollution data are only available after the monitoring program, we obtain ambient air quality measures from AOD via the Moderate Resolution Imaging Spectroradiometer (MODIS) algorithm, operated by NASA and installed on the platform of the research satellite, *Terra*. The original data have a geographic resolution of 10-by-10 km and a scanning frequency of 30 minutes, which we average to the city-week level from 2011 to 2016. MODIS records the degree to which sunlight is scattered or absorbed in the entire atmospheric column corresponding to the overpassed area under cloud-clear condition. As such, AOD captures concentration of particles such as sulfates, nitrates, black carbons, and sea salts, and serve as a proxy for outdoor particulate matter pollution (Van Donkelaar, Martin, and Park 2006). Online Appendix Figure E.5 documents a strong correspondence between AOD and PM_{2.5} after the implementation of the monitoring program.¹⁷

II. Information Access and Awareness

In this section, we document how the monitoring program improved public access to pollution information and increased consumer awareness of air pollution. Unless noted otherwise, for each outcome we report (i) a national-level time series graph from January 2011 to April 2016, and (ii) an event study graph at the city level from eight quarters before the monitoring program reached the city and five quarters after.¹⁸ Both graphs display raw data patterns with no control variables included, though the patterns are essentially the same with spatial or temporal controls as shown in online Appendix Figure E.6. The coefficient for event quarter -1 (the quarter prior to the monitoring program) is normalized to 0. We report the 95 percent confidence interval constructed from standard errors clustered at the city level.

A. Information Access: News and Mobile Apps

News media and mobile phone apps are among the primary venues for the general public to access pollution information. In panel A of Figure 3, we count the number of days in each month when the *People's Daily* mentions “smog” in any articles. “Smog” is rarely mentioned in the 1990s and 2000s. Almost immediately following the monitoring program’s initial rollout, the frequency of “smog” appearing in the *People's Daily* jumped from two days per month in 2012 to sixteen days per month in 2013. It is possible that the initial increase in media coverage may have

¹⁷ An important advantage of the MODIS AOD measure is that it is available at high temporal frequency, which allows us to exploit weekly variation in air pollution to identify short-term responses of avoidance behavior and health outcomes. Most other remote-sensing measures of air pollution, such as the widely used Van Donkelaar et al. (2016), are provided at the annual level. MODIS AOD also uses a direct retrieval algorithm that infers air quality purely through optical properties of sunlight reflection and absorption; that MODIS AOD is free from complicated atmospheric and chemistry modeling helps alleviate concerns that the measurement itself may change over time in ways that might be correlated with the policy change. For example, MODIS AOD is not calibrated to ground PM_{2.5} monitoring data and therefore the measurement itself is not influenced by the change in the availability of PM_{2.5} data because of China’s monitoring policy.

¹⁸ The event window is chosen to provide the longest balanced panel possible given the data available. The data covers the period from January 2011 to April 2016, and the wave 1/2/3 cities received pollution information in January 2013/November 2013/December 2014, respectively. This allows us to examine 8/11/16 prepolicy quarters and 13/10/5 post-policy quarters for these cities.

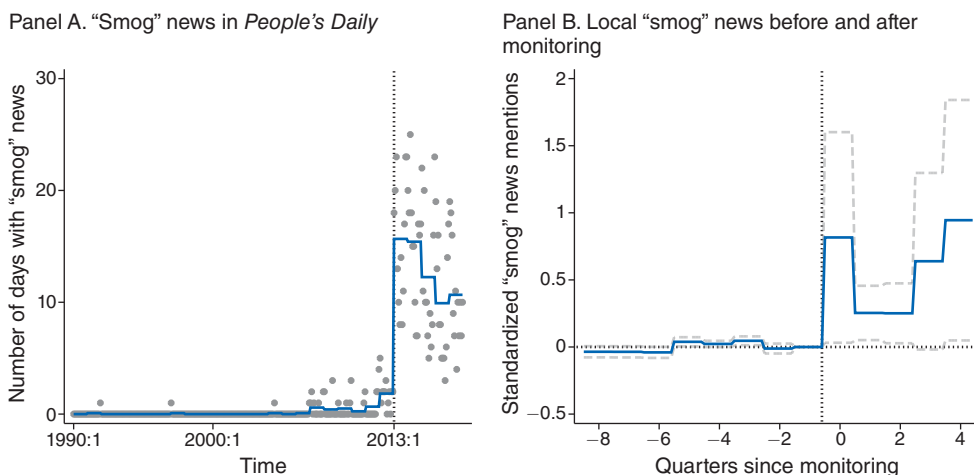


FIGURE 3. CHANGES IN POLLUTION INFORMATION ACCESS: NEWS MENTIONS

Notes: Panel A plots the number of days in each month when the *People's Daily*, the official newspaper of the Chinese government which reports both national and local news, published articles containing "smog" in content. Each dot represents one month. Line shows annual averages. Panel B plots standardized city-level "smog" mentions, defined as news that mentions both "smog" and the city name, as a function of quarters since the completion of the monitoring program in a given city. Event quarter -1 is normalized to 0. The underlying regression contains no control variables; see online Appendix Figure E.6 for versions with fixed effects controls. Dashed lines show 95 percent confidence interval constructed from standard errors clustered at the city level.

been due to coverage of the monitoring program itself. However, it is unlikely that this is the sole reason for the sustained high level of media coverage throughout the rest of the sample period.¹⁹

The national trend in media's smog mention may reflect changes in the general environment (e.g., shifts in government policies). To hone in on the impact of the monitoring program, we scan each smog-containing article in the *People's Daily* to determine the list of cities mentioned. This allows us to construct a city panel dataset and, combined with the staggered rollout of the monitoring program, conduct an event study for changes in frequency of smog mentions before and after a city begins to monitor pollution.

Panel B of Figure 3 provides an event-time plot of smog mentions as a function of quarters relative to a local city's monitoring program rollout, where smog mention is standardized to zero mean and unit variance. The graphical pattern features a discrete increase exactly on the rollout date (event quarter $t = 0$). By one year after the rollout, smog mentions in cities with monitoring stations increased by 60 percent of a standard deviation (p -value = 0.038). Other keywords, such as "air pollution" and "atmospheric pollution," exhibited similar increases in media reports after the monitoring program.

Next, we examine the availability of mobile phone apps, which can provide users with real-time updates on air quality information, in contrast to newspapers that only provide daily updates. We compare the number of newly released pollution apps

¹⁹There was a modest increase in the frequency of the term "smog" before 2013. A close read of the articles indicates that these phrases were mostly used to describe dense fog and were rarely associated with pollution.

with that of other popular app categories, including gaming, music, video, reading, finance, sports, education, shopping, and navigation. As shown in online Appendix Figure E.7, there is a clear surge in the number of pollution apps released after the monitoring program, relative to nonpollution apps. Among all pollution apps that were available for download in December 2015 (i.e., when we queried for the data), 82 percent were released between 2013 and 2015, compared to 18 percent that were released between 2009 and 2013. In contrast, 38 percent and 62 percent of the apps in the control group were released during 2009–2013 and 2013–2015, respectively.

B. Awareness: Web Searches

We measure the demand for pollution-related information by Internet keyword searches on Baidu. This analysis is analogous to the examination of “smog” news in Section IIA. Again, we focus on search queries for “smog,” though the patterns for keywords “air pollution,” “mask,” and “air purifier” are very similar.

Figure 4 repeats the exercise in Figure 3 but uses the search index as the outcome variable. Panel A of Figure 4 plots the time-series pattern of the search index for “smog” at the national level. The index varies between 0 and 200 for most of 2011 and 2012 and jumps overnight from 175 in December 2012 to 4,400 in January 2013, the month of the initial rollout. This is a 25-fold increase. The search index also began to exhibit a strong seasonal pattern after 2013, with the highest values occurring during the winter seasons. This mirrors the seasonality of smog, which tends to be more severe during the winter months due to the increased use of coal for heating.²⁰ The search indices of 21,000 in December 2013 and 25,000 in January 2015, two outliers that were omitted from the graph for readability, correspond to the two worst smog episodes in the country’s recent history.

Leveraging disaggregated search indices for over 300 cities, panel B of Figure 4 provides an event-time plot of the standardized search indexes two years before and one year after the rollout in a local city. Echoing results in Section IIA, the search index remains flat before the monitoring program and rises rapidly after monitoring starts. Within one year after the rollout, smog searches increase by on average 75 percent of a standard deviation.²¹

The surge in Internet searches both nationally and locally and their strong correlation with air pollution after the start of the monitoring program provide compelling evidence that concepts such as “smog” and “air pollution,” as well as related adverse health consequences, entered the public domain as a result of easy access to information provided by the monitoring program. As forcefully put by Wainwright (2014) and quoted by Greenstone and Schwarz (2018), “(by 2014,) the daily talk of the AQI has become a national pastime amongst ex-pats and Chinese locals alike. Air quality apps are the staple of every smartphone. Chinese micro blogs and parenting

²⁰ We report correlation coefficient estimates in online Appendix Table E.2.

²¹ As pollution tends to peak during the winter months in China, we expect web searches to increase immediately after the monitoring rollout as well as 9–12 months after the rollout (the following winter). This is indeed what we find: media mentions of smog (panel B of Figure 3), web searches (panel B of Figure 4), and air purifier purchases (panel B of Figure 7) all exhibit similar patterns. Importantly, there were no spikes in these metrics 9–12 months before the monitoring rollout, indicating a lack of awareness in the absence of pollution information.

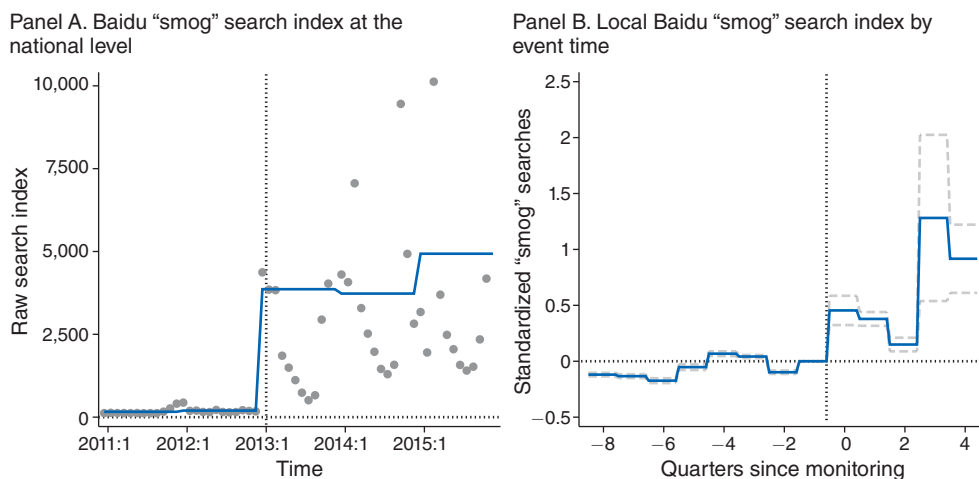


FIGURE 4. CHANGES IN POLLUTION AWARENESS: BAIDU SMOG SEARCH INDEX

Notes: Panel A plots raw monthly trends in Baidu Search Index for keyword “smog.” The graph omits two dots with exceptionally high search index (December 2013 with index = 20,942 and December 2015 with index = 24,679) for readability. Line shows annual averages. Panel B plots standardized “smog” search index as a function of quarters since the completion of the monitoring program in a given city. Event quarter –1 is normalized to 0. The underlying regression contains no control variables; see online Appendix Figure E.6 for versions with fixed effects controls. Dashed lines show 95 percent confidence interval constructed from standard errors clustered at the city level.

forums are monopolized by discussions about the best air filters and chatter about holidays to ‘clean air destinations.’”

C. Awareness: Public Opinion

Public opinion toward environmental issues has also changed as a result of the monitoring program. Our analysis uses data from the China Family Panel Studies (CFPS), a nationally representative, bi-annual longitudinal survey of Chinese families and individuals. In 2012, the CFPS began including a set of questions that ask respondents to rate the perceived level of concern for various social and economic issues, including corruption, education, environment, housing, inequality, jobs, medical care, and social security, on a scale from 1 to 10. For our analysis, we utilize the CFPS data from the 2012, 2014, 2016, and 2018 waves, comprising 83,237 respondents living in 126 cities. Of these cities, 34, 39, and 49 are in the first, second, and third wave of the monitoring program rollout, respectively.

We investigate whether the monitoring program influenced concerns about the environment and other subjects by regressing the respondents’ expressed level-of-concern metric on a post-monitoring dummy, while controlling for respondent fixed effects and year fixed effects.²² Panel A of Figure 5 shows the changes

²²Specifically, we estimate the following regression equation:

$$Concern_{icy}^s = \beta^s \cdot Post_{cy} + \alpha_i^s + \alpha_y^s + \varepsilon_{icy}^s,$$

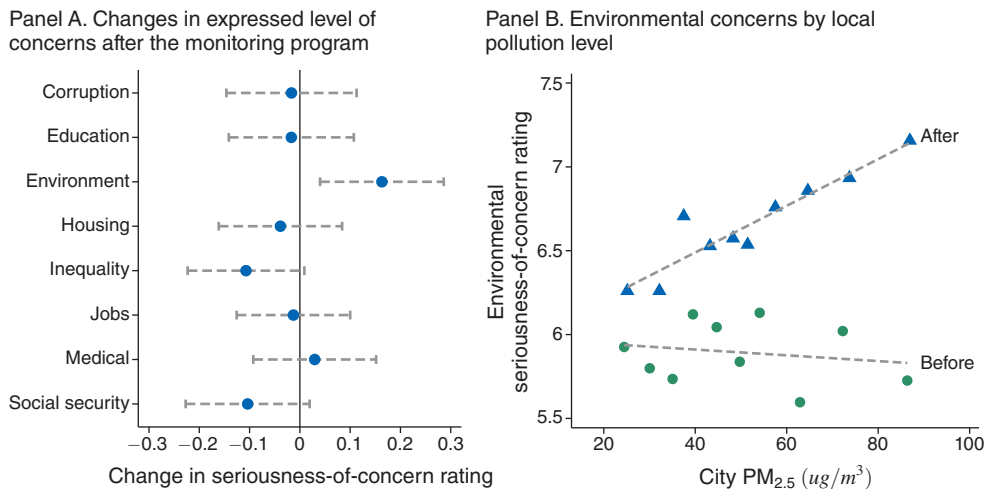


FIGURE 5. CHANGES IN PUBLIC ATTITUDES TOWARD ENVIRONMENTAL, ECONOMICS, AND SOCIAL ISSUES

Notes: Panel A examines whether the monitoring program changed the public's attitudes towards various subjects, regressing respondents' level-of-concern metric on a post-monitoring dummy, controlling for respondent fixed effects and year fixed effects. Each bar corresponds to a separate regression, showing the coefficient estimate on the post-monitoring dummy and the 95 percent confidence interval. Panel B displays bin scatterplots between respondents' level-of-concern on environmental issues and their local city's 2015–2016 average PM_{2.5} concentration, separately by before and after the monitoring program.

in the level of concern for all eight subjects. The results indicate that respondents had increased concerns about environmental issues after the monitoring program. In contrast, there were no statistically significant changes in respondents' views towards other economic and social issues.

In panel B of Figure 5, we further show that the increase in respondents' environmental concerns after monitoring policies are driven primarily by those live in cities with high levels of air pollution. Specifically, we relate a respondent's environmental concern with his/her city's 2015–2016 average PM_{2.5} concentration.²³ We find that respondents' concerns about environmental issues are essentially uncorrelated (or slightly negatively correlated) with their cities' pollution levels before the monitoring program (shown by the circles); environmental concerns and pollution become strongly positively correlated after monitoring (shown by the triangles). That is, individuals living in cities with worse air quality become disproportionately more concerned with pollution than individuals living in clean cities.

where $\{s, i, c, y\}$ denotes $\{\text{subject}, \text{individual}, \text{city}, \text{year}\}$, and the equation is estimated separately for each subject s . The dependent variable Concern_{icy}^s represents the level of concern expressed by individual i about subject s on a scale of 1–10. The key regressor Post_{cy} is an indicator for city-years after the pollution information rollout. We assign 2014, 2016, and 2018 as the post period for cities in waves 1 and 2 of the monitoring rollout, and 2016 and 2018 as the post period for cities in wave 3. α_i^s and α_y^s are individual and year fixed effects, respectively. We weight all regressions by the individual-level survey weight to obtain nationally representative estimates and cluster standard errors at the city level.

²³ Cities in different information rollout waves have different monitoring start dates. In particular, wave 3 cities did not start monitoring until the end of 2014. We therefore measure city's average pollution using 2015–2016 data so that pollution levels are consistently calculated based on the same time period for all cities.

To provide additional support for our findings, we also analyze data from the “message board” for government leaders, a government web portal that allows citizens to leave anonymous comments, including complaints and questions, to local government officials (Agarwal et al. 2020). Our data includes over 1.3 million comments from 2010 to 2019. We use this data to count the number of air pollution-related comments (those that include “smog,” “PM_{2.5},” “air pollution,” or “air quality” in the text) over time. For comparison, we also look at comments related to water pollution issues (comments with “water pollution” or “water quality” in the content). Online Appendix Figure E.8 plots these trends. Air pollution related complaints rose sharply in 2013, which coincided with the launch of the monitoring program. No similar spike was observed for water pollution comments.

III. Empirical Framework

Our results so far demonstrate that the monitoring program significantly expanded access to pollution information and elevated households’ awareness and concerns towards pollution issues. These changes may in turn influence the behavior of individuals who take steps to reduce the negative effects of pollution. In online Appendix B, we formalize this idea using a stylized model that incorporates the elements of information economics (Hirshleifer 1971; Hilton 1981) into a classical model of health demand and production (Grossman 1972; Harrington and Portney 1987). In particular, we show that an information program that reduces the wedge between individuals’ perceived and experienced air pollution can (i) increase pollution avoidance behavior and (ii) reduce health damage from the same amount of outdoor pollution. The model also provides a framework to estimate the costs and benefits of the monitoring program which we come back to in Section VI. In this section, we describe the econometric implementation of these theoretical predictions.

Our empirical objective is to examine whether the monitoring program has changed *avoidance behavior and health effects of pollution*. In the literature, these terms are typically measured using “pollution gradients,” i.e., $\partial \text{Outdoor Activities} / \partial \text{Pollution}$ for pollution avoidance and $\partial \text{Health} / \partial \text{Pollution}$ for pollution’s health damage.²⁴ Our innovation to the literature is to evaluate how these pollution gradients change as the monitoring program drastically improves the public access to pollution information.

A. Econometric Specification

We estimate the changes in the outcome-pollution gradient using the following equation:

$$(1) \quad \text{Outcome}_{ct} = \alpha \text{Pollution}_{ct} + \beta \text{Pollution}_{ct} \times \text{Post}_{ct} + \mathbf{x}'_{ct} \boldsymbol{\gamma} + \varepsilon_{ct}$$

²⁴Examples include Graff Zivin and Neidell (2009), Deschenes, Greenstone, and Shapiro (2017), and Deryugina et al. (2019) which study the effects of pollution on outdoor activities, defensive drug spending, and mortality, respectively.

where c denotes a city and t denotes week. $Pollution_{ct}$ is a measure of ambient air pollution that equals the log of AOD in the city-week. The dummy variable $Post_{ct}$ represents the information treatment, and takes the value one for all periods after city c implements the monitoring program. $\mathbf{x}_{ct}$ is a vector of control variables including the $Post_{ct}$ term, city fixed effects, year fixed effects, week-of-year fixed effects, and city-specific time trends.²⁵ Unless otherwise stated, all regression analyses and event studies control for these fixed effects. In robust specifications, we include more flexible controls such as week-of-sample effects and region-by-week-of-sample effects. ε_{ct} is the error term. We cluster standard errors at the city level to allow for arbitrary serial correlations among the sample periods. In equation (1), α represents the outcome-pollution gradient before the monitoring program, and β , the primary coefficient of interest, denotes changes in the gradient after the program. We defer the identifying assumptions underlying the causal inference of β to Section IIIB.

For each outcome, we also implement an event study version of equation (1) by replacing the $Post_{ct}$ dummy with a series of quarter-since-monitoring indicators:

$$(2) \text{ Outcome}_{ct} = \alpha \text{ Pollution}_{ct} + \sum_{q=-8}^k \beta_q \text{ Pollution}_{ct} \times d(t \in q)_{ct} + \mathbf{x}'_{ct} \boldsymbol{\gamma} + \varepsilon_{ct},$$

where $d(t \in q)_{ct}$ indicates week t being in the q th quarter since monitoring began in city c .²⁶ We plot the estimates of β_q 's to provide a visual representation of how the relationship between pollution and outcome, i.e., the outcome-pollution gradient, evolves as a function of time relative to monitoring rollout. Cities in different waves have different numbers of pre- and post-periods. Our event studies trace out the evolution of β_q 's starting eight quarters before the monitoring rollout. Depending on data availability (Table 1), we report post-monitoring β_q 's for five quarters in the analysis of purchase trips and air-purifier sales ($k = 5$) and eight quarters in the analysis of mortality ($k = 8$). These event time windows are chosen to ensure a balanced underlying sample for each coefficient. We dummy out sample periods outside of the event window. We use the same estimation framework as described in equations (1) and (2) throughout the regression analysis below.

Equations (1) and (2) highlight the difference between our study and the previous literature that estimates the causal effect of air pollution exposure. Conventionally, the key threat to identification arises because pollution exposure is likely to be correlated with the error term: $E[\text{Pollution}_{ct} \times \varepsilon_{ct}] \neq 0$. Such endogeneity could be due to omitted variables or measurement errors in pollution exposure. The scope of our empirical analysis differs in that, in most cases, we are not interested in the causal effect of pollution per se (which is α), but rather in the *change* in the causal effect that takes place after the monitoring program (which is β). The key insight of our empirical framework is that, under reasonable assumptions, one can consistently

²⁵ City fixed effects control for all permanent difference of city characteristics in the cross section. City-specific time trends account for any differential trends in the outcome that may be related to the policy being studied, such as the differential rate of card penetration across cities in different waves. Controlling for city-specific trends is demanding and results in more conservative estimates (see Barwick et al. 2019, the original NBER Working Paper version of this paper, that does not include time trends controls). In unreported analysis, we have confirmed that other specifications of time trends, such as quadratic or cubic trends, or trends that are allowed to vary by pre- and post-monitoring periods, all produce similar results.

²⁶ We report quarterly (rather than weekly) event study coefficients to average out noise in the trends. This does not affect the estimation of the average program impact.

estimate the change in pollution's causal effects (β) using OLS, without having to consistently estimate the level of the effect (α). In Section IIIB that follows, we discuss identification issues in more details.

B. Identification

We now describe the identification assumptions required to causally estimate β using OLS estimation equation (1). Intuitively, if we were to separately estimate the outcome-pollution slope using data before and after the treatment, the endogeneity in pollution would lead to inconsistency in both estimates. However, if the nature of the endogeneity is not affected by the treatment, the inconsistency in the slope estimates could cancel out, leaving the OLS estimate of β consistent. The following two assumptions formalize this intuition:

IDENTIFICATION ASSUMPTION 1: $\varepsilon_{ct} \perp Post_{ct} | \mathbf{x}_{ct}$. *This standard exogeneity assumption implies that conditioning on city attributes and other controls $\mathbf{x}_{ct}$ (e.g., city and time fixed effects, and city-specific time trends), the treatment status $Post_{ct}$ is exogenous. That is, the program rollout is as good as random conditional on the control variables.*

IDENTIFICATION ASSUMPTION 2: $Post_{ct} \perp Pollution_{ct} | \mathbf{x}_{ct}$. *This assumption implies that the treatment status is independent of pollution levels conditioning on $\mathbf{x}_{ct}$ (i.e., variations in pollution unexplained by $\mathbf{x}_{ct}$). The fact that the first-wave cities tend to have higher pollution levels on average does not violate this assumption per se as the average pollution levels can be controlled by city fixed effects. This assumption requires the treatment status being exogenous to the pollution variation unexplained by $\mathbf{x}_{ct}$. It ensures that the nature of the endogeneity in pollution does not change before and after the policy.²⁷*

We provide formal econometric arguments in online Appendix C. Intuitively, there are two sources of inconsistency in the OLS estimate of β in equation (1): one from smearing due to the endogeneity of $Pollution_{ct}$ and the other from the endogeneity of the interaction term $Pollution_{ct} \times Post_{ct}$. We show that inconsistency from these two sources cancels out—therefore leaving the OLS estimate of the difference (the change in the pollution gradient that we are interested in) to be consistent—under the two identification assumptions. Section IIIC next presents empirical support of two identification assumptions and discusses additional empirical challenges.

C. Potential Identification Threats

The identification of α and β , the parameters of interest in equation (1), is based on the monitoring program's staggered rollout across cities. The two key identification assumptions outline above imply that the rollout schedule is not correlated with time-varying unobservables that could affect the outcome variables or pollution

²⁷ This assumption is stronger than what we need to prove the consistency of β as shown in online Appendix C.

levels *conditioning on the controls*. Here we first examine the validity of these two assumptions. We also discuss potential estimation biases with a staggered rollout and adjustments for multiple-hypothesis testing—as we examine multiple outcomes and subgroup heterogeneity—to reflect the best practices that are proposed in recent literature.

City Assignment to Monitoring Rollout Waves.—As the assignment of cities to different monitoring rollout waves is not random, here we investigate the consequences of this selection for our econometric analysis. First, the rollout wave assignment mainly corresponds to administrative hierarchies that are determined long before the policy, with a few exceptions due to exogenous factors such as geographic contiguity (Section IA). We present a visualization in online Appendix Figure E.9. Colors indicate administrative hierarchies: deep blues are cities in the Jing-Jin-Ji Metropolitan Region, the Yangtze River Delta Economic Zone, the Pearl River Delta Metropolitan Region, direct-administered municipalities, and provincial capitals; light blues are 2007 Environmental Improvement Priority Cities and 1997–2007 National Environmental Protection Exemplary Cities; whites are all other cities. If rollout follows *strictly* administrative hierarchies, then all deep blue cities should be in wave 1, all light blue cities should be in wave 2, and all white cities should be in wave 3. This is true 93 percent, 88 percent, and 86 percent of the time for wave 1, 2, and 3. Among the exceptions, we calculate that 55 percent, 69 percent, and 100 percent of the cases (or 68 percent of total cases) are due to the fact that they border cities in the corresponding wave and it makes more sense administratively to group them into those waves.

Second, following Almond, Li, and Zhang (2019), in online Appendix Section D.1, we present a more formal, “zero-stage” analysis on the determinants of city rollout assignment. The idea is that, if the rollout wave assignment is mostly driven by predetermined administrative hierarchies and status, then conditional on these characteristics, one should expect little explanatory power from additional controls, such as pollution or economic conditions. Results in online Appendix Table E.3 confirm this. Conditioning on administrative hierarchies, levels especially trends in cities’ prepolicy air pollution or income provide little to no additional explanatory power for city wave assignment, consistent with Identification Assumptions 1 and 2 of Section IIIB.

Third, online Appendix Table E.4 summarizes economics and environmental attributes for cities in each wave. Cities in earlier waves tend to have a larger population, higher income, and higher levels of air pollution and industrial emissions. We note that our regression analyses include city fixed effects, and therefore permanent differences in city characteristics do not pose threats to our identification. In online Appendix Table E.5, we further test whether time-varying city characteristics correlate with policy rollout after controlling for the suite of fixed effects controls we use in equation (1). We examine a rich set of characteristics including pollution levels, local political and regulatory environment (e.g., the number of local officials ousted during the anti-corruption campaign, demographics of local political leaders, news mentions of regulation policies), and healthcare access (the number of medical facilities). Conditioning on the suite of fixed effects, the program rollout is not correlated with these city attributes. These “balancing checks” suggest a lack

of city-level confounders that correlate with the timing of policy rollout, supporting Identification Assumptions 1 and 2 of Section IIIB.

Finally, the event study as outlined in equation (2) helps diagnose potential endogeneity in rollout timing through preexisting trends in the outcome of interest. The event studies in Figures 6 and 8 provide further support for Identification Assumption 1 of Section IIIB.

Concurrent Policies as Potential Confounders.—Since 2014, the Chinese government has implemented a variety of environmental and energy programs as part of its efforts to combat air pollution, as noted in Greenstone et al. (2021). Here, we discuss several major policies and their potential to interfere with our analysis of the impact of the monitoring program. First, China rolled out pilot CO₂ cap-and-trade programs targeted at heavily polluting industries in two provinces (Guangdong in December 2013 and Hubei in April 2014) and five individual cities (Shenzhen in June 2013, Beijing and Shanghai in November 2013, Tianjin in December 2013, and Chongqing in June 2014).²⁸ The second policy is the energy reduction plan among over 16,000 of the largest energy users that collectively accounted for over 60 percent of national energy consumption in 2010. The policy started in 2012 and aimed to reduce energy intensity and carbon intensity as outlined in the twelfth Five-Year Plan. The third policy is fuel switching from coal to natural gas for winter heating in northern China from the winter season of 2013, as part of the *Action Plan on Air Pollution Prevention and Control*.

Because the geography and timeline of these concurrent environmental policies do not significantly overlap with the monitoring program that we study in this paper, we expect the impact of these policies to be captured by the extensive set of spatial and temporal controls in our econometric models (e.g., city fixed effects and region-by-week-of-sample fixed effects). As shown in online Appendix Table E.5, the monitoring program is not associated with any significant changes in pollution levels after controlling for fixed effects. This is supportive evidence, as if these other environmental policies were systematic confounders, we would expect our model to show an improvement in air quality due to the monitoring program.²⁹

Bias in Staggered DID.—Recent literature points out that the two-way fixed effects model in a staggered difference-in-differences (DID) setting could yield biased estimates when there are heterogeneous treatment effects over time or across units (De Chaisemartin and d'Haultfoeuille 2020a,b; Sun and Abraham 2021; Callaway and Sant'Anna 2021; Goodman-Bacon 2021). One reason for this is that in two-way fixed effects DID models, treated units that are treated earlier serve as “controls” for later treated units. However, if there are heterogeneous treatment effects across units or over time, the treatment can cause units to be on different

²⁸These seven regional pilots were a proof of concept for the national CO₂ cap-and-trade program slated to come online in 2020. However, the program had a limited impact on carbon emission and air quality given the generous allocation and the lack of a strong punishment mechanism for noncompliance (Zhang, Wang, and Du 2017).

²⁹It is important to clarify that our findings are not contradictory to the fact that China's *overall* air quality has significantly improved since 2015, due in part to various environmental regulations mentioned above (e.g., Greenstone et al. 2021). These trends are absorbed in our flexible city and time fixed effects. Our results show that, conditional on these fixed effects, the monitoring policy itself does not have an independent effect on improving air quality.

trends. This can make the post-treatment periods for earlier treated units no longer a valid comparison for later treated units, potentially leading to biased estimates.

Our research differs from previous studies in that we focus on the impact of the monitoring program on economic- and health-pollution *gradients*, rather than on the outcomes themselves. As a result, the standard solutions proposed in the literature do not directly apply to our study. However, to evaluate the potential for bias, we follow the intuition outlined in Goodman-Bacon (2021) and assess the robustness of our findings by restricting comparisons to “switcher” units and “not-yet-treated” units. Because our study involves three rollout waves, there are a limited number of such comparisons. We examine each of them individually and report robustness results in Sections IVD and VC.

Multiple Testing Adjustment.—In both of our analyses on card transactions and mortality, we are interested in subgroup heterogeneity – for example, whether the monitoring program generates differential impacts for deferrable versus nondeferrable transactions or the elderly versus nonelderly population. Since we examine multiple outcomes, we control for the family-wise error rate, i.e., the probability of incorrectly rejecting one or more null belonging to a family of hypotheses. Following the best practices as recommended in Aguilar-Gomez et al. (2022), we report the family-wise adjusted p -values for the β coefficients based on 100 bootstraps of the free step-down procedure of Westfall and Young (1993), with standard errors clustered at the city level. Specifically, we first define an exclusive family of hypotheses for both card transactions and mortality. Each family contains the overall category and all subgroup categories.³⁰ We implement the Westfall-Young procedure using the algorithm provided by Jones, Molitor, and Reif (2019) and conduct the block bootstraps simulations at the city level.

Causal Effect of Pollution.—The main goal of our study is to estimate how the monitoring program changes the pollution gradient. Our key econometric insight is that, under reasonable and testable assumptions, this parameter can be causally identified by estimating the β coefficient in equation (1) using simple OLS. An alternative approach is to estimate the causal impacts of air pollution separately for periods before versus after the implementation of the monitoring program. We report this analysis in Section VC using instrumental variables (IV) estimation strategies from the literature on the health effects of air pollution.

IV. Pollution Information and Behavioral Responses

A. Avoidance Behavior

We first examine how the monitoring program changes household avoidance behavior. We use the bank card transaction data to build a measure of shopping trips, defined as the number of transactions in city c at week t per 10,000 active cards. As we explained in Section IB, we focus on transactions that take place

³⁰In other words, we define a family as a column in Tables 2, 3, and 4. For example, for card transactions, a family is a column in Table 2 that includes the overall, deferrable, and nondeferrable transactions.

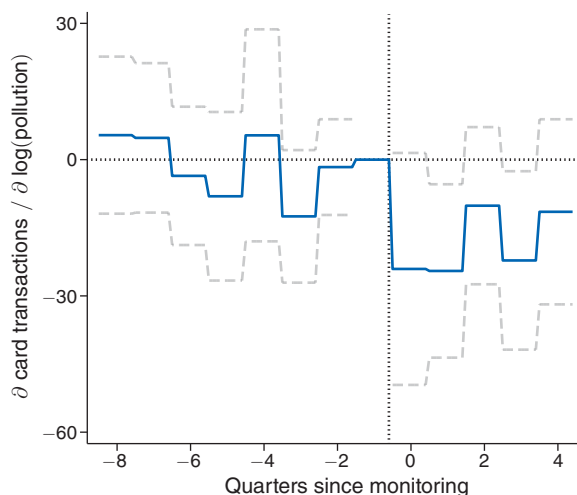


FIGURE 6. CHANGES IN AVOIDANCE BEHAVIOR: SHOPPING-POLLUTION GRADIENT

Notes: The figure shows coefficients from a regression of city-weekly band card transaction frequencies per 10,000 active cards on satellite-based pollution (in logarithm) as a function of quarters since the completion of the monitoring program. Since all transactions in our data occurred at physical stores, these transactions can be used as a proxy for outdoor trips. The shopping-pollution gradient, therefore, serves as a measure of avoidance behavior. Event quarter -1 is normalized to 0. Each line segment represents a coefficient estimate. The underlying regression controls for city, week-of-year, and year fixed effects, and city-specific time trends. Dashed lines show 95 percent confidence interval constructed from standard errors clustered at the city level.

in physical stores, which enables us to track outdoor activities. We use this measure of shopping trips as the outcome variable in equation (1), and therefore the β coefficient represents the change in the shopping trip-pollution gradient $\partial \text{Outdoor Activities} / \partial \text{Pollution}$ due to the monitoring program, reflecting avoidance behavior.

Figure 6 shows an event study graph for β_q 's, with the coefficient for the event quarter -1 (β_{-1}) normalized to zero. Two patterns emerge. First, before the program, the β_q 's estimates are flat and also statistically indistinguishable from zero, suggesting a lack of pretrends. Second, the purchase-pollution gradient became negative almost immediately after the rollout of the monitoring program. Doubling the pollution readings leads to about 11 fewer trips per 10,000 cards on a weekly basis.

We examine the statistical precision and the robustness of these patterns in panel A of Table 2. Column 1 controls for city, week-of-year, and year fixed effects, as well as city-specific time trends, which correspond to the specification of Figure 6. Column 2 also includes week-of-sample fixed effects (fixed effects for all weeks during 2011–2016), exploiting variation in pollution across cities in the same week. Column 3 further adds region-by-year fixed effects, allowing for common trends in transactions and pollution that are specific to each region.³¹ Column 4 is our most stringent specification, controlling for region-by-week-of-sample fixed effects in addition to city fixed effects and city trends. We obtain similar results across the

³¹“Region” is a conventional partition of cities by location: north (36 cities), northeast (38 cities), east (105 cities), central south (81 cities), southwest (54 cities), and northwest (52 cities).

TABLE 2—CHANGES IN AVOIDANCE BEHAVIOR: SHOPPING-POLLUTION GRADIENT

	(1)	(2)	(3)	(4)
<i>Panel A. All transactions</i>				
$\log(\text{Pollution})$	5.78 (3.72)	3.74 (4.08)	2.38 (3.37)	4.75 (3.55)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	-10.50 (4.45) [0.048]	-13.30 (5.19) [0.032]	-8.22 (4.59) [0.136]	-12.17 (5.24) [0.056]
<i>Panel B. Merchant type: deferrable (supermarket, dining, entertainment)</i>				
$\log(\text{Pollution})$	3.77 (1.50)	3.31 (1.77)	2.87 (1.38)	3.69 (1.65)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	-8.33 (1.57) [0.000]	-9.02 (2.22) [0.00]	-6.51 (2.04) [0.0140]	-8.69 (2.31) [0.010]
<i>Panel C. Merchant type: nondeferrable (all other categories)</i>				
$\log(\text{Pollution})$	2.01 (2.53)	0.43 (2.65)	-0.50 (2.28)	1.06 (2.32)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	-2.17 (3.24) [0.536]	-4.29 (3.35) [0.228]	-1.71 (2.98) [0.598]	-3.48 (3.53) [0.400]
FEs: city, city-linear time trends	✓	✓	✓	✓
FEs: week-of-year	✓			
FEs: year	✓			
FEs: week-of-sample		✓	✓	
FEs: region $\times$ year			✓	
FEs: region $\times$ week-of-sample				✓

Notes: Number of observations is 83,122. “ $\log(\text{Pollution})$ ” is logged AOD in a city $\times$ week. Dependent variable is city $\times$ weekly bank card transactions per 10,000 active cards. Mean of the dependent variable is 869.1 (panel A), 314.4 (panel B), and 554.8 (panel C). “Region” is a conventional partition of cities by location: north (36 cities), northeast (38 cities), east (105 cities), central south (81 cities), southwest (54 cities), and northwest (52 cities). Standard errors clustered at the city level reported in parentheses. Family-wise adjusted p -values based on the step-down resampling procedure of Westfall and Young (1993) are in brackets.

board. The parameter estimate on the interaction term suggests that when pollution doubles, weekly card transactions would decrease after the program by 12 more per 10,000 cards than before the program, representing a 1.4 percentage points decrease from the mean transaction frequency of 869.1 transactions per 10,000 cards. In other words, the shopping-pollution elasticity becomes 1.4 percentage points more negative post the monitoring program. A simple derivation shows that the program reduced card transactions by 1.4 percent, which we will use to calculate the welfare loss from foregone consumption in the cost-benefit analysis.

The observed effect is economically significant. As a point of reference, Cutter and Neidell (2009) find that when a “Spare the Air” alert is issued in the San Francisco Bay Area, *daily* traffic falls by 2.5–3.5 percent with the largest effect during and just after the morning commuting period.³² Graff Zivin and Neidell (2009) estimate that a *one-day* smog alert issued in Southern California leads to an 8–15 percent reduction in attendance at two major outdoor facilities (the Los

³²The “Spare the Air” (STA) advisories, designed to elicit voluntary reductions in vehicle usage and encourage the usage of public transit and carpooling, are issued on days when ground-level ozone is predicted to exceed National Ambient Air Quality Standards.

Angeles Zoo and the Griffith Park Observatory) on the same day, though the effect dissipates quickly in consecutive days of such alerts. While these two studies focus on daily behavioral changes after government-issued air quality warnings, our analysis focuses on changes in shopping behavior at the weekly level, which naturally accounts for within-week intertemporal substitution. Additionally, our data covers *all* consumption categories and represents 59 percent of national consumption.

Table 2 also shows that shopping trips are not affected by pollution before the program, as indicated by the insignificant coefficients for the main “log(Pollution)” term. While our purchase-pollution slope estimate prior to the monitoring program may not be causal (as previously discussed), this finding is consistent with the idea that households did not take effective mitigating measures in the absence of information about pollution levels. We discuss this issue in more detail in Section IVC below.

Subcategory Spending.—We expect pollution avoidance behavior to be more relevant for consumption trips that can be reasonably delayed. In panels B and C of Table 2, we examine heterogeneity between “deferrable” and “nondeferrable” consumption. We define deferrable consumption as transactions that occurred in supermarkets, restaurant dining, and entertainment merchants. As expected, the monitoring program generates a steeper response among deferrable transactions where doubling pollution leads to a 2.6 percentage points reduction (column 4), which explains about 74 percent of the change in the overall shopping-pollution gradient. The estimated effects on the rest of the consumption categories (“nondeferrable”) are negative but not statistically significant.

This pattern of subgroup heterogeneity stands up to multiple hypothesis testing adjustments. As explained in Section IIIC, we used the method described in Westfall and Young (1993) to calculate family-wise adjusted p -values, which are reported in brackets in Table 2. The results suggest both responses of overall and deferrable transactions are statistically significant.

While we have classified shopping trips as deferrable or nondeferrable in an effort to identify the categories most affected, this classification may seem somewhat arbitrary. In online Appendix D.2, we describe an alternative, data-driven method to identify transactions that are more “outdoor” in nature. We reiterate that our analysis only includes transactions that occur in physical stores, which by definition involve some level of outdoor activities. However, there are some types of spending activities, such as visiting recreation parks, that may involve a longer time spent outside and potentially more pollution exposure compared to activities like shopping in malls. Specifically, we examine how precipitation shocks affect transaction activities across 273 narrowly defined merchant categories. The idea is that categories that are sensitive to precipitation events may be more likely to involve outdoor activities or be more easily deferred, and therefore may also be the most affected categories when individuals are trying to avoid air pollution. We relegate full details to online Appendix D.2 (with results summarized in online Appendix Figure E.10). Here we highlight several key findings.

First, the data-driven exercise appears to detect deferrable and/or outdoor trips successfully. The most precipitation-sensitive categories include snow car rentals and dealers, vacation houses, sports stadiums, amusement parks, resort areas, and

garment stores, among others that can potentially be postponed or typically take place in open-air settings.³³ Second, we conducted the pollution avoidance analysis again using shopping trips from the top 20 most precipitation-sensitive transaction categories as the dependent variable. Our results show that doubling the pollution concentration would lead to 5.5 percent fewer transactions after the monitoring rollout, indicating a larger response compared to the overall consumption gradient change we observed in panel A of Table 2.

B. *Defensive Spending*

In addition to reducing outdoor trips during times of high pollution, households may also increase “defensive” spending that directly reduces their pollution exposure. We examine purchases of air purifiers, which are popular household appliances in China used to mitigate the effects of severe air pollution (Ito and Zhang 2020).

We obtain monthly sales of air purifiers for 50 large cities in China between 2012 and 2016 (Section IB). Panel A of Figure 7 plots the raw monthly time series of total units sold. Air purifier sales more than doubled and rose from 11,000 units per month in 2012 to over 25,000 units per month after 2013. The change in air purifier sales patterns appears to mirror the changes in web search behavior shown in Figure 4, with strong seasonality in air purifier sales emerging after 2013, often with large spikes in sales during the winter months. At the city level, similar to the pattern we see in web searches, the increase in air purifier sales coincides with the timing of the monitoring program rollout and doubles within four quarters of the rollout (panel B of Figure 7). Further evidence that the relationship between pollution and air purifier sales (the purchase-pollution gradient) became stronger after the rollout of the monitoring program is provided in online Appendix Table E.6, which shows that sales became more positively correlated with month-over-month variations in air pollution after the availability of pollution information.

C. *Private Information and Avoidance*

The monitoring program resulted in the widespread dissemination of *public information* about air pollution. Individuals may also use *private information* to respond to pollution, such as visual or olfactory cues. In this subsection, we investigate the role of two sources of private information (air visibility and data from the US Embassy) in pollution avoidance. In particular, we examine whether these sources of private information helped households avoid pollution before the monitoring program was implemented. A summary of our findings is provided here, with further details in online Appendices D.3 and D.4.

The Role of Air Visibility.—In online Appendix D.3, we assess the possibility that people used air visibility as a proxy for pollution levels before the pollution information program. We report three main sets of findings. First, panel B in online

³³ Additionally, it is reassuring to find that transactions for supermarkets, restaurant dining, and entertainment—which we identified as deferrable in Table 2, panel B—are disproportionately deferrable and outdoor in nature from a precipitation-sensitivity perspective.

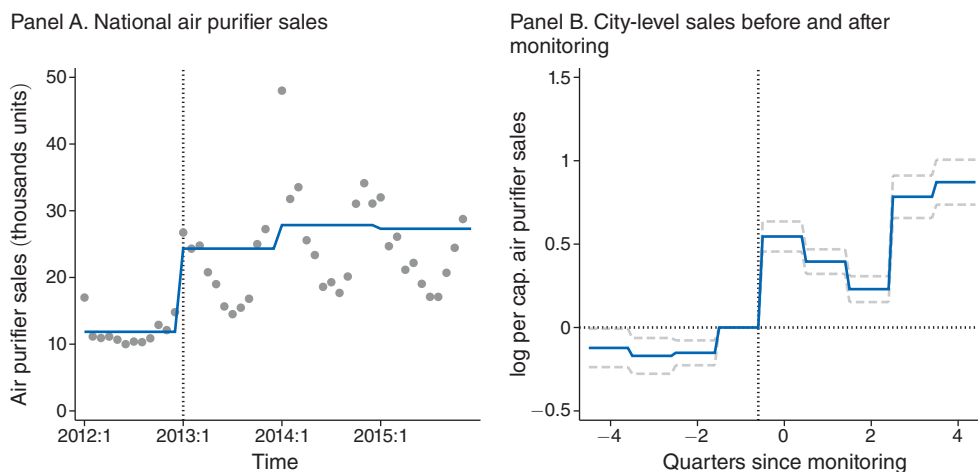


FIGURE 7. CHANGES IN DEFENSIVE SPENDING: AIR PURIFIER SALES IN 50 CITIES

Notes: Panel A plots raw monthly trends in national air purifier sales from offline venues. The graph omits two dots with exceptionally high sales for readability. These dots correspond to December 2013 (sales = 61,605 units) and December 2015 (sales = 74,352 units). Line shows annual averages. Panel B plots log per capita air purifier sales as a function of quarters since the completion of the monitoring program in a given city. Event quarter -1 is normalized to 0. The underlying regression contains no control variables; see online Appendix Figure E.6 for versions with fixed effects controls. We report four quarters of pretreatment coefficients because the air purifier data are available since January 2012, one year prior to the initial rollout of the monitoring program. Dashed lines show 95 percent confidence interval constructed from standard errors clustered at the city level.

Appendix Figure E.12 shows that while visibility is correlated with ambient pollution levels, the relationship has a large spread and is particularly noisy when visibility is poor. The pattern underlies challenges that people face in trying to avoid pollution without actual pollution monitoring data: both heavy fog and heavy smog could cause poor visibility outdoors.

Next, we reestimate equation (1) and use poor visibility as the key right-hand-side variable instead of pollution. Panel C in online Appendix Figure E.12 shows that before the monitoring program, there were few card transactions during weeks with the visibility in the lowest quintiles. However, the shopping-visibility relationship is nonmonotone, economically modest, and imprecisely estimated. The reduction in card transactions during weeks with the lowest visibility may reflect a genuine concern for poor visibility per se (e.g., road safety concerns) instead of pollution.

More importantly, we find that adding actual pollution (measured with AODs) as control variables has little impact on the estimated shopping-visibility relationship before the program as shown in panel C in online Appendix Figure E.12. If people were using visibility as an indicator of *pollution* levels, we would expect the relationship between shopping and visibility to change once we directly control the pollution levels in the regression. This is indeed what we observe *after* the monitoring program as shown in panel D of online Appendix Figure E.12: the negative relationship between poor visibility and transactions attenuates once we control for pollution. These findings suggest that households were unlikely to have been using visibility as a means of effectively avoiding pollution before the monitoring program.

US Embassy Data.—The US Embassy in Beijing installed a rooftop air quality monitor in April 2008 before the upcoming summer Olympics and started reporting hourly $PM_{2.5}$ levels via its Twitter account (@beijingair) to advise US citizens who travel and do business in China. United States consulates in Chengdu, Guangzhou, Shanghai, and Shenyang followed the practice in later years. Among the five cities with US Embassy and Consulates, Beijing is the only city with independent $PM_{2.5}$ data dating back to 2011. We therefore focus our analysis on Beijing and modify equation (1) to test to examine whether the $PM_{2.5}$ monitoring conducted by the United States was an effective source of information for pollution avoidance by local residents (see online Appendix Section D.4 for more details).

Online Appendix Figure E.13 summarizes the results. Both weekly web searches for smog and card transactions exhibited a precisely estimated, close-to-zero correlation with embassy-reported air pollution before 2013. Analogous to our main findings, after 2013, the correlation between web searches and embassy-reported pollution becomes significantly positive, and the correlation between card transactions and pollution becomes significantly negative. The fact that the correlations were close to zero before 2013 and then increased significantly after 2013 suggests that the US Embassy $PM_{2.5}$ data were not a significant source of information for Beijing residents prior to 2013, when public $PM_{2.5}$ data were not available. This is likely because Twitter was not accessible in China and the embassy data were not widely distributed among the general public, despite attracting attention from environmental activists.³⁴

D. Additional Robustness Checks

Robustness Specifications.—Online Appendix Table E.7 examines the robustness to sample selections, alternative variable definitions, and additional controls. The first row of online Appendix Table E.7 excludes the top 10 percent of the cities with the highest number of officials who were ousted as a result of China's sweeping anti-corruption campaign, which has been associated with a reduction in luxury consumption (Qian and Wen 2015). The second row adds controls for the share of card transactions that occurred through online venues, as people might substitute offline trips with online purchases. The third row adds weather controls, including weekly average temperature, precipitation, wind speed, barometric pressure, and their interactions. The fourth row replaces the weekly average AOD readings with the maximum daily AOD value observed during the week. In the fifth row, we redefine the pre and post monitoring periods based on the "field" rollout dates that we manually collected from news media and city government yearbooks (Section IA). Online Appendix Figure E.14 plots the distribution of these dates. We do not use these field dates in our preferred specification because there may be significant measurement errors, such as delays or rounding of dates in news reporting. For example, many cities in wave 3 describe the start of the monitoring program generally as "starting

³⁴ US Embassy's pollution data can serve as important information source where residents have access to it. Jha and Nauze (2022) show that the installation of US Embassy air pollution monitoring in 50 cities around the world and the real-time Tweets of air quality information led to increases in local pollution awareness, as captured by Google Searches for terms related to air pollution.

in 2015,” while the actual start date is likely in December 2014 according to the policy deadline. Lastly, we aggregate the data and conduct analysis at the quarterly frequency to capture avoidance effects that may manifest throughout a longer time horizon and to account for potential intertemporal substitution of consumption across different weeks of the quarter.

These robustness tests produce estimates that are similar across the board and generally indistinguishable from one another. The quarterly analysis on the last row produces a larger effect estimate than would be implied by a linear scale-up of the weekly estimate. This suggests pollution avoidance may extend beyond a weekly horizon, although some of the quarterly estimates are imprecise due to a 13-fold reduction in the sample size.

Spatial Spillovers.—Online Appendix Table E.8 examines potential spatial spillover effects, e.g., whether new coverage about pollution episodes in one city raises awareness in nearby cities that do not have pollution monitoring yet. For each wave of treated cities, we introduce a group of neighboring cities that share borders with the treated cities but have not installed monitoring stations. These neighboring cities are assigned the same rollout time as their treated counterparts. We allow the changes in the shopping-pollution gradient to differ between the treated and neighboring cities.³⁵ There are two key findings. First, there are no detectable changes in the shopping-pollution gradient in neighboring cities at the time when the monitoring program began in the treated cities. Second, the shopping-pollution gradient change in treatment cities relative to that of their neighbors is similar in magnitude to the main effect reported in Table 2, though the triple interaction term is estimated with less precision than the main specification. The lack of spatial spillover of one city’s information rollout to its neighbor cities suggests that having actual data about *local* pollution levels is important in shaping avoidance behavior.

Staggered Rollout.—As discussed in Section IIIC, to assess the potential bias due to staggered treatment, we follow the intuition laid out in Goodman-Bacon (2021) and only compare treated cities and not-yet-treated cities. Specifically, we conduct three separate analyses and compare cities in wave one with cities in the latter two waves, cities in wave one with cities in wave three, and cities in wave two with cities in wave three. We only include data from periods when the control group cities have not yet started pollution monitoring. Panel A of online Appendix Figure E.15 illustrates the sample cities and periods for each analysis. The point estimates in panel B of online Appendix Figure E.15 vary across columns (for example, it is smaller in the specification that compares wave 2 cities with wave 3 cities), though primarily due to differences in the means of the dependent variable. The shopping-pollution gradient using different comparison groups ranges from 1 percent to 1.8 percent, similar to our baseline findings.

³⁵ We incorporate two interaction terms: “ $\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$ ” that captures changes in neighboring cities and “ $\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\} \times \mathbf{1}\{\text{Treated}\}$ ” that capture additional changes in treated cities. Cities in the last wave of the rollout are excluded from the analysis. Including these cities as treated units without neighboring pairs produces similar results.

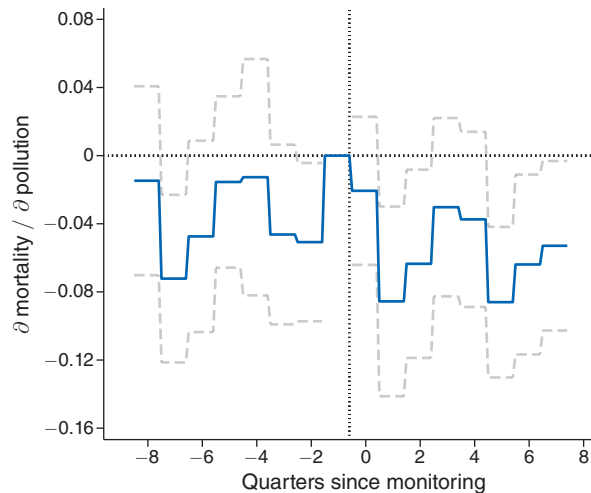


FIGURE 8. CHANGES IN HEALTH DAMAGE: MORTALITY-POLLUTION GRADIENT

Notes: This figure shows coefficients from a regression of log mortality rate on log satellite-based pollution as a function of quarters since the completion of the monitoring program in a given city. Event quarter -1 is normalized to 0. Each line segment represents a coefficient estimate. The underlying regression controls for city, week-of-year, and year fixed effects, and city-specific time trends. Dashed lines show 95 percent confidence interval constructed from standard errors clustered at the city level.

V. Pollution Information and Mortality Benefit

A. Mortality Responses

Our previous analyses have documented changes in pollution information and awareness as well as a range of behavioral responses to the monitoring program. Our endpoint analysis examines whether the same amount of pollution exposure is associated with fewer deaths after pollution information becomes widely available as a result of risk-compensating behavioral responses. Our analysis uses weekly mortality data in 131 cities from 2011 to 2016.

Figure 8 plots the β_q coefficient estimates for the event study described by equation (2), where the dependent variable is the log of mortality rates. The mortality-pollution elasticity exhibits a flat trend before the program, followed by a noticeable decline afterward, consistent with the event study analysis on behavioral changes above. This decline appears to strengthen after a few quarters, suggesting that the mortality effect tends to manifest over time. Table 3 reports the mortality analysis using equation (1), with the same specifications as in Table 2. Panel A examines the mortality of all causes. The monitoring program leads to a significant 1.3 to 2.3 percentage points reduction in the mortality-pollution elasticity. That is, pollution became less deadly after the program: doubling pollution is associated with 1.3 to 2.3 percent fewer deaths after the program.

TABLE 3—CHANGES IN THE HEALTH DAMAGE: MORTALITY-POLLUTION GRADIENT

	(1)	(2)	(3)	(4)
<i>Panel A. General mortality</i>				
$\log(\text{Pollution})$	0.0069 (0.0062)	0.0059 (0.0068)	0.0039 (0.0049)	0.0020 (0.0057)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	-0.0228 (0.0069) [0.006]	-0.0190 (0.0076) [0.044]	-0.0159 (0.0064) [0.038]	-0.0126 (0.0068) [0.134]
<i>Panel B. Causes of death: cardiorespiratory</i>				
$\log(\text{Pollution})$	0.0113 (0.0070)	0.0104 (0.0075)	0.0097 (0.0062)	0.0091 (0.0072)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	-0.0312 (0.0079) [0.002]	-0.0257 (0.0083) [0.020]	-0.0254 (0.0074) [0.010]	-0.0238 (0.0083) [0.022]
<i>Panel C. Causes of death: noncardiorespiratory</i>				
$\log(\text{Pollution})$	-0.0003 (0.0067)	-0.0018 (0.0073)	-0.0043 (0.0061)	-0.0086 (0.0071)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	-0.0130 (0.0088) [0.246]	-0.0096 (0.0095) [0.372]	-0.0052 (0.0088) [0.560]	-0.0012 (0.0100) [0.902]
<i>Panel D. Causes of death: injury</i>				
$\log(\text{Pollution})$	-0.0132 (0.0080)	-0.0172 (0.0085)	-0.0191 (0.0082)	-0.0209 (0.0097)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	0.0002 (0.0089) [0.990]	0.0125 (0.0100) [0.372]	0.0169 (0.0101) [0.178]	0.0204 (0.0115) [0.142]
FEs: city, city-linear time trends	✓	✓	✓	✓
FEs: week-of-year	✓			
FEs: year	✓			
FEs: week-of-sample		✓	✓	
FEs: region $\times$ year			✓	
FEs: region $\times$ week-of-sample				✓

Notes: Numbers of observations are: 36,369 (panel A), 36,226 (panel B), 36,262 (panel C), and 32,621 (panel D). “ $\log(\text{Pollution})$ ” is logged AOD in a city $\times$ week. Dependent variable is city $\times$ weekly logged mortality rate. “Region” is a conventional partition of cities by location: north (36 cities), northeast (38 cities), east (105 cities), central south (81 cities), southwest (54 cities), and northwest (52 cities). Standard errors clustered at the city level are reported in parentheses. Family-wise adjusted p -values based on the step-down resampling procedure of Westfall and Young (1993) are in brackets.

B. Heterogeneity

Cause- and Age-Specific Mortality Responses.—We continue our mortality analysis by examining subgroup responses. Panels B to D of Table 3 breaks down overall mortality by the underlying causes of death. Panel B focuses on cardiorespiratory mortality and shows a 2.4 to 3.1 percentage point decrease in the mortality-pollution elasticity. Panel C shows that the effect on noncardiorespiratory mortality is smaller and statistically insignificant. During our study period, cardiorespiratory deaths account for about one-half of total deaths. These estimates imply that about 80 percent of the reduction in mortality damage after the monitoring program comes from cardiovascular and respiratory causes which are widely considered as the most direct consequences of pollution exposure. Panel D performs a placebo test that examines

mortality that is associated with external injuries. There is no evidence that the monitoring program reduces injury-related mortality, as expected.

Table 4 reports the mortality effect for four age groups: infants, age 1–14, age 15–59, and age 60 and above.³⁶ The effect of the monitoring program concentrates among the preteen and the elderly who are more vulnerable to pollution exposure than mid-aged adults. Interestingly, we find no change in the mortality-pollution relationship for infants under one year of age. This could be associated with the low pollution exposure among infants due to the traditional Chinese practice of keeping newborns indoors within the first few months of their birth to minimize outdoor exposure.

In Table 3 and Table 4, the numbers in brackets report the Westfall and Young (1993) family-wise adjusted p -values using standard errors clustered at the city level. For each table, a family of hypotheses includes the outcomes in all panels for each of the four types of fixed effects specifications. The coefficient of interest “ $\log(Pollution) \times 1_{\{after\ monitoring\}}$ ” survives multiple testing adjustment at the 5 percent significance level for cardiorespiratory mortality (Table 3, panel B) and for elderly mortality (Table 4, panel D). The family-adjusted p -value for the all-cause mortality effect (Table 3, panel A) varies from 0.006 to 0.134.

Mortality Responses and City Characteristics.—We further conduct heterogeneity analysis which provides suggestive evidence on the mechanism underlying the mortality effect. Specifically, we repeatedly split the sample into two using the average value of a series of city-level characteristics, including per capita income, share of the urban population, per capita number of hospitals, per capita residential electricity use, and share of mobile phone users. Results in panel B of online Appendix Table E.11 suggest a larger reduction in mortality damage in cities that experience a higher mobile phone penetration, consume a higher rate of residential electricity, and have more hospitals. Cities that have higher per capita income and are more urban also report a larger mortality reduction, though the estimates are noisier. These findings are consistent with the observation that residents in above-median cities are better able to engage in defensive spending to reduce the health damage from air pollution exposure, and hence benefit more from pollution information.

In panel A of online Appendix Table E.11, we repeat the heterogeneity analysis for shopping trips using the same set of city attributes. The overall patterns of the two panels appear consistent: residents in above-median cities exhibit stronger (though imprecisely estimated) avoidance behavior to elevated pollution. While there are various ways to protect oneself from the harmful effects of pollution, the data suggests that avoiding outdoor activities may be a short-term strategy for reducing the harm caused by pollution.

C. Robustness Checks

Instrumental Variables Approach.—Our econometric arguments in Section IIIB and online Appendix C show that one can causally estimate *changes* in the effects

³⁶The low number of observations for the infant and preteen analysis (panels A and B of Table 4) is due to city-weeks with zero deaths.

TABLE 4—CHANGES IN THE HEALTH DAMAGE: AGE-SPECIFIC MORTALITY-POLLUTION GRADIENT

	(1)	(2)	(3)	(4)
<i>Panel A. Age groups: infants</i>				
$\log(\text{Pollution})$	0.0104 (0.0074)	0.0097 (0.0074)	0.0032 (0.0075)	0.0033 (0.0097)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	−0.0044 (0.0116) [0.766]	−0.0052 (0.0130) [0.918]	0.0070 (0.0137) [0.692]	0.0009 (0.0161) [0.950]
<i>Panel B. Age group: 1–14</i>				
$\log(\text{Pollution})$	0.0218 (0.0141)	0.0206 (0.0145)	0.0161 (0.0135)	0.0092 (0.0118)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	−0.0338 (0.0136) [0.068]	−0.0304 (0.0127) [0.062]	−0.0255 (0.0124) [0.130]	−0.0196 (0.0134) [0.336]
<i>Panel C. Age group: 15–59</i>				
$\log(\text{Pollution})$	−0.0051 (0.0063)	−0.0108 (0.0071)	−0.0125 (0.0060)	−0.0166 (0.0069)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	−0.0055 (0.0082) [0.766]	0.0038 (0.0093) [0.918]	0.0070 (0.0091) [0.692]	0.0145 (0.0094) [0.336]
<i>Panel D. Age group: 60 and above</i>				
$\log(\text{Pollution})$	0.0106 (0.0053)	0.0104 (0.0057)	0.0100 (0.0050)	0.0082 (0.0058)
$\log(\text{Pollution}) \times \mathbf{1}\{\text{after monitoring}\}$	−0.0316 (0.0065) [0.000]	−0.0271 (0.0068) [0.002]	−0.0264 (0.0059) [0.002]	−0.0229 (0.0067) [0.008]
FEs: city	✓	✓	✓	✓
FEs: week-of-year	✓			
FEs: year	✓			
FEs: week-of-sample		✓	✓	
FEs: region $\times$ year			✓	
FEs: region $\times$ week-of-sample				✓

Notes: Numbers of observations are: 14,060 (panel A), 12,626 (panel B), 35,932 (panel C), and 36,272 (panel D). “ $\log(\text{Pollution})$ ” is logged AOD in a city $\times$ week. Dependent variable is city $\times$ weekly logged mortality rate. “Region” is a conventional partition of cities by location: north (36 cities), northeast (38 cities), east (105 cities), central south (81 cities), southwest (54 cities), and northwest (52 cities). Standard errors clustered at the city level are reported in parentheses. Family-wise adjusted p -values based on the step-down resampling procedure of Westfall and Young (1993) are in brackets.

of pollution on outcomes (β in equation (1)) using simple OLS under reasonable identification assumptions, without having to causally identify the *level* of the pollution’s effects (α in equation (1)). This said, the levels of the effects, both before and after the monitoring rollout, could nevertheless be of academic and policy interest.

We present an instrumental variables (IV) strategy where we isolate plausibly exogenous variation in a city’s local air pollution that is attributable to transported pollutants from upwind cities (e.g., Barwick et al. 2020). We then estimate two-stage least squares (2SLS) regressions of mortality on air pollution, instrumenting pollution with the IV variable. We run the 2SLS regressions separately for periods before monitoring and after monitoring.

Online Appendix D.5 discusses in detail the IV construction and empirical results. Reassuringly, the changes in the mortality impact of air pollution (the mortality-pollution gradient) before and after the program are virtually the same

whether the estimates are based on OLS or from IV regressions. This evidence supports the key identification assumption underlying the change-in-gradient analysis of equation (1). The analysis also produces causal estimates on pollution's mortality impact for different underlying causes of death, separately for before and after the monitoring program.

Synthetic Control Analysis.—An alternative way to assess the impact of the monitoring policy is to examine changes in mortality outcomes directly, rather than changes in mortality-pollution gradients. If changes in pollution's effects are large enough across the spectrum of pollution levels, one could in principle detect an overall reduction in mortality rates. The key to this exercise is to identify a set of “control” cities for each treated city that can serve as reasonable counterfactuals in the absence of the monitoring rollout. We adopt the synthetic control method proposed by Abadie, Diamond, and Hainmueller (2010) and Arkhangelsky et al. (2021). Following best practice, we pair each treated city with control cities that have not yet been exposed to the monitoring program and exclude wave three cities (those in the last wave without “not-yet-treated” controls). We calculate synthetic weights using data up to one year prior to the rollout of information and use the latter half of the pretreatment period as a validation sample to evaluate the synthetic control method. We summarize our findings here and relegate details to online Appendix Section D.6.

Panel A of online Appendix Figure E.18 shows an event study for the synthetic analysis. Several patterns emerge. First, there are no obvious trends in the mortality differences between treated cities and synthetic control cities prior to the monitoring rollout, which validates the synthetic analysis. Second, relative to control cities, treated cities' mortality rates decrease by 1.64 percent on average after the monitoring program, though the drop is imprecisely estimated. Third, the reduction in mortality rates seems to be more concentrated during heavily polluted periods (highlighted by the gray areas on the chart) following the introduction of the monitoring program, a pattern that was not previously apparent. These patterns suggest that the imprecise average effect of the monitoring program may be masked by important heterogeneity.

Panel B of online Appendix Figure E.18 explores this heterogeneity more formally by allowing the monitoring program's mortality effect to differ across quintiles of pollution levels. The program has no discernible impact on the mortality rate during periods of low pollution levels, but reduced mortality significantly (both statistically and economically) during high pollution periods. For example, the mortality rate decreases by nearly 4 percent in weeks with the 20 percent highest pollution levels. Importantly, this result is isomorphic to our gradient analysis in Section VA, where we document a negative coefficient for the pollution and post-treatment interaction. In theory, this negative relationship documented in Section VA could be driven by an increase in mortality during low pollution periods, a decrease in mortality during high pollution periods, or some combination of both. The analysis here shows that our main finding of the monitoring program's mortality benefit is mostly driven by a net reduction in mortality during high pollution periods, further justifying our focus on the gradient analysis which leverages the fact that the mortality impact of the program is critically through its interaction with pollution. That is, rather than a

(uniform) level shift in the mortality rate across different pollution levels, the program resulted in a change in the slope of the mortality-pollution relationship.

Staggered Rollout.—In panel B of online Appendix Figure E.15, we compare the mortality rates of treated cities only with those of not-yet-treated cities. The results show that the estimated mortality impact is similar to the baseline when we compare wave one cities with waves two and three or wave three alone, and slightly larger when we compare wave two cities with wave three. This suggests that our main estimate of a 1.3 to 2.3 percentage point reduction in the mortality-pollution elasticity (as reported in Table 3) may be conservative.

VI. Cost-Benefit Analysis of the Monitoring Program

We end with a cost-benefit analysis of the monitoring program that quantifies the benefit and cost components outlined in equation (B.2) in the online Appendix theory section. As this exercise depends on several assumptions and estimates that we borrow from the literature, we try to be conservative in the benefit estimation while more liberal in the cost estimation.

A. Benefits of the Monitoring Program

Mortality Benefit.—Together with our estimates on the change in mortality-pollution gradient in Section V, the mortality benefit is quantified using age-adjusted value of statistical life (VSL). The VSL approach is commonly used by policymakers to evaluate the benefits of life-saving regulations. Due to the lack of reliable VSL estimates for the Chinese population, we use a benefit transfer method that infers the VSL for Chinese residents from US-based VSL estimates and the income elasticity of VSL. The benefit-transfer method is commonly used for countries with insufficient or unreliable data (Viscusi and Masterman 2017).

Ashenfelter and Greenstone (2004) estimate that the VSL in the United States is \$2.3 million, which can be considered as a lower bound.³⁷ Narain and Sall (2016) suggest a transfer elasticity of 1.2 for transferring the VSL from the United States to a developing country. Combined with a one-to-nine income ratio between China and United States, the VSL for population in China amounts to \$1.3 million in 2015.³⁸ Based on the age adjustment suggested by Murphy and Topel (2006), the average age-adjusted VSL is estimated to be 0.49 million for individuals 60 and above, among whom the mortality impact is concentrated. The β estimate in column 4 and panel D in Table 4 suggest that the program has reduced total mortality for the age group 60 and above by at least 2.3 percentage points, amounting to a savings of 114,330 deaths per year.³⁹ Therefore, the mortality benefit from reduced premature deaths among people 60 and above is 56.2 billion (about \$9 billion) annually.

³⁷The US EPA recommends a central VSL estimate of US\$(2006)7.4 million, or US(2015)\$8.9 based on a meta analysis of VSL studies.

³⁸The VSL for China = $\$2.3 / (1.2 \times 9)$ million = \$0.21 million, or 1.3 million.

³⁹The total number of deaths among people 60 and above is 4.971 million in 2012 according to China Statistical Yearbook.

Morbidity Benefit.—The behavioral changes that reduce premature deaths could also lower morbidity. Relying on comprehensive data on health spending, recent studies such as Deschenes, Greenstone, and Shapiro (2017) and Barwick et al. (2020) have shown that the healthcare cost alone, not accounting for the quality of life impact, is about two-thirds of the mortality cost in United States and China, respectively. This would imply a morbidity benefit of about 37.1 billion (\$5.9 billion) per year.

The combined mortality and morbidity benefits amount to at least 93.3 billion (\$14.9 billion) annually. This likely reflects a lower-bound of the program's benefit for several reasons. First, we use a conservative VSL estimate of \$2.3 million for the United States. Second, the benefit calculation does not account for the quality-of-life changes or productivity gains from improved health. Third, to be consistent with the analysis on other outcomes, the mortality-pollution relationship is estimated on weekly data, and hence short-run in nature. In an analysis based on quarterly data, we find the longer-run reduction in the mortality-pollution gradient is indeed slightly larger (more lives saved than reported in Table 4).

B. Costs of the Program

We now turn to the costs of avoidance, which include increased defensive spending, the welfare loss from foregone consumption, and the cost of the monitoring program itself. First, we consider purchases of air purifiers and masks as defensive expenditures. Aggregate annual sales of air purifiers and facial masks grew from 4 billion in 2012 to 13.5 billion in 2015. The average increase in annual sales amount to 7.5 billion during 2013–2015, relative to 2012.⁴⁰ One caveat in this calculation is that a complete cost calculation of defensive spending would necessitate information on drug uses, changes in residential locations, as well as other nonmarket activities (such as outdoor exercises), which we do not consider in this paper.

Second, to calculate the welfare loss from foregone consumption, we use estimates from Table 2 to calculate foregone consumption and assume the price elasticity for consumer goods to vary from -1 to -1.5 (Deaton and Muellbauer 1980; Blundell, Pashardes, and Weber 1993; Banks, Blundell, and Lewbel 1997). The estimated deadweight loss from the foregone consumption ranges between 1.3 and 1.9 billion.⁴¹ The estimated foregone consumption are upper bounds as some of them are likely deferred rather than permanently foregone: as shown in Section IVA, the effect on bank-card use appears to concentrate on “deferrable” categories, many of which are temporally substitutable in nature.

Third, the one-time cost to set up the monitoring stations and broadcast the pollution information online is estimated to be at most 2 billion, and the annual operation

⁴⁰For air purifier sales, see “Industry Trends for China’s Air Purifiers,” by Growth for Knowledge (2015); for facial mask sales, see “PM_{2.5} Facial Masks Industry Report,” by Intelligence Research Group (2017).

⁴¹The deadweight loss can be approximated by $1/2 \times \Delta \text{consumption} \times \tau/|\epsilon|$, where $\Delta \text{consumption}$ is the change annual in consumption (in ¥) due to the program; τ is the ratio of the change in consumption over total consumption; and ϵ is the price elasticity for consumption. The β estimate in column 4 and panel A in Table 2 suggests that τ is 1.4 percent (12.17 transactions out of 869 per 10,000 cards), and that $\Delta \text{consumption}$ amounts to 278 billion.

and maintenance cost is at most 0.1 billion a year.⁴² Taken together, a likely upper bound of the total costs is 11.5 billion in its first year (including the set-up costs) and 9.5 billion annually thereafter.

The estimated annual benefits and costs of the program (at least 93.3 billion of health benefits relative to the associated costs of 9.5 billion) underscores the cost-effectiveness of the information monitoring and disclosure program.⁴³ Empowering the public with real-time pollution information mobilizes individuals' ability to mitigate the adverse consequences of pollution at a relatively low cost. This, together with other environmental regulations, can more effectively reduce the societal cost of pollution.

VII. Conclusions

This paper examines the role of pollution information in enhancing households' awareness of pollution problems, promoting pollution avoidance behaviors, and mitigating pollution's negative health impacts. We focus on a landmark policy change in China that resulted in the installation of air pollution monitoring stations nationwide and the provision to the public of the real-time pollution information that previously had been unavailable. Based on several rich and unique datasets, our analysis provides strong evidence that the program resulted in cascading changes such as increased access to pollution information, enhanced awareness of pollution's harm, more pronounced avoidance behavior, and higher defensive spending. The risk-compensating behavioral changes ultimately led to a reduction in the negative health effects of air pollution. Our estimates suggest that the health benefits of the information monitoring and disclosure program is nearly one order of magnitude larger than the combined costs of the program implementation itself and associated behavioral changes.

China's experience offers an important lesson for other developing countries that are facing severe environmental challenges. In many low-income countries the infrastructure for monitoring environmental quality and disclosing the information is often inadequate or does not exist. As income rises, the demand for environmental quality increases, and households are better able to adapt to adverse environmental conditions. Providing real-time pollution data, combined with effective dissemination infrastructure such as smartphones and the Internet, both of which are increasingly available in developing countries, could be a powerful and cost-effective tool to help households mitigate health damages from environmental pollution. Such information transparency and distribution may also enhance the effectiveness of other environmental regulations and related enforcement mechanisms (Greenstone et al. 2022) that are designed to reduce air pollution. Furthermore, the optimal design of the monitoring system (a subject that in itself warrants more research attention) should help provide more accurate air quality measurements and thus help unleash

⁴²Chinese central and local governments invested 1.8 billion in setting up the nationwide monitoring system (Wang 2014). The winning bids for the operation and maintenance of all the monitors amount to about 213 million from 2016 to 2018 according to from the Chinese government procurement information platform.

⁴³Successful environmental regulations such as the US Acid Rain Program and the Clean Air Act have been shown to achieve benefits an order of magnitude larger than the cost (Krupnick and Morgenstern 2002; Chestnut and Mills 2005). Different from our context, these programs substantially reduced pollution levels.

the full extent of health benefits that can accrue through the provision of accurate, actionable information.

Finally, while the finding of large health benefits from air quality monitoring and disclosure program is consistent with the notion that information remedies can serve as a quasi-regulatory mechanism (Konar and Cohen 1997), our findings do not suggest that such information programs can or should serve as a substitute for policies that aim to reduce the dangerous levels of air pollution in many developing countries. The dramatic improvement in air quality in China during the last few years likely would not have occurred without a series of dramatic and stringent policies to reduce emissions (Greenstone et al. 2021; Karplus, Zhang, and Zhao 2021). By facilitating better enforcement and civic engagement, the monitoring program might have enhanced the effectiveness of environmental regulations to improve air quality. We consider this an important question for future research.

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